Host connection
UriSed

Authorization

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Contents

1. Purpose

The purpose of this document is to specify the possible communication protocol types of the UriSed sediment urine analyzer.

2. Scope

This interface description applies only for the user software of the UriSed project. The sales of the UriSed project are: UriSed, UriSed 2, UriSed 2 FL, UriSed 3, UriSed 3 PRO.

3. Interface Summary

This chapter describes the different physical connections.

3.1 Physical interfaces

	Comment
Serial	connects to host via RS-232 port
Ethernet	connects to host via LAN

3.1.1 RS-232 interface settings

The following configuration of the RS-232 serial communication protocol settings are present in the user software:

- Baud rate:
  - 9 600
  - 14 400
  - 19 200
  - 38 400
  - 57 600
  - 115 200

Non-configurable settings in the user software:

- Length:
  - 8
- Parity check:
  - N (no)
- Stopbits:
  - 1

3.1.2 Ethernet

If the Ethernet is used, the target computer's data must be provided to open the communication channel. Always LIS is the server and UriSed is the client. The following configuration of the LAN must be provided:

- IP address
- Port

4. Unidirectional communication with host computer - 'Unidirectional up to 1.6.1'

4.1 Low Level Protocol between UriSed and host computer:

STX + [frame] + ETX

STX: Start of Text
ETX: End of Text

4.2 Frame definitions:

4.2.1 Structure of the frame for both chemical and sediment result

```
URINE ANALYZIS REPORT[CR][LF]
[CR][LF]
[CR][LF]
Laboratory :[SP28][Labname][CR][LF]
[CR][LF]
Sample ID :[SP29][Sample ID][CR][LF]
Patient name :[SP26][Patient name][CR][LF]
Date&Time :[SP29][Date&Time][CR][LF]
General chemical result :[SP15][General chemical result][CR][LF]
General sediment result :[SP15][General sediment result][CR][LF]
N° of + particle types :[SP16][N° of + particle types][CR][LF]
N° of images :[SP26][N° of images][CR][LF]
[CR][LF]
CHEMISTRY RESULT[CR][LF]
pad[SP17]Conv.[SP16]Arbitr.[SP13][CR][LF]
[PAD_NAME1][CH20][SI_Result1/CONV_Result1][CH20][ARB_Result1][CH20][CR][LF]
[PAD_NAME2][CH20][SI_Result2/CONV_Result2][CH20][ARB_Result2][CH20][CR][LF]
…
[PAD_NAMEN][CH20][SI_ResultN/CONV_ResultN][CH20][ARB_ResultN][CH20][CR][LF]
[CR][LF]
SEDIMENT RESULT[CR][LF]
[CR][LF]
[SP21]p/µl[SP16]N°[SP18]+/-[SP17]Ref. (p/µl)[SP9][CR][LF]
[PARTICLE_NAME1][CH20][p/µl_Result1][CH20][N°_Result1][CH20][+/-_Result1][CH20][Ref.
(p/µl)_Result1][ SP1][CR][LF]
[PARTICLE_NAME2][CH20][p/µl_Result2][CH20][N°_Result2][CH20][+/-_Result2][CH20][Ref.
(p/µl)_Result2][ SP1][CR][LF]
…
[PARTICLE_NAMEN][CH20][p/µl_ResultN][CH20][N°_ResultN][CH20][+/-_ResultN][CH20][Ref.
(p/µl)_ResultN][ SP1][CR][LF]
```

```
[CR][LF]
Comment : [Comment][CR][LF]
Dilution factor : [Dilution factor][CR][LF]
```

In case there is no chemical result, then that part of the transfer is not visible (there is no 'CHEMISTRY RESULT' title and table)

In case there is no sediment result, then 'SEDIMENT RESULT - Invalid measurement' title is visible, there is no sediment table, and 'Not sediment measured' comment is visible.

The result display only SI or Conv unit depending from the settings. In the sediment part the other unit name will be No. And the result value will be empty string "".

4.2.1.1 Chemical (LabUMat2) and sediment connected result example:

```
[STX]URINE ANALYZIS REPORT


Laboratory :                        Please fill it in SETTINGS/Maintenance menu !

Sample ID :                         0121
Patient name :                      -
Date&Time :                         07/07/2021 09:50:42
General chemical result :           +
General sediment result :           -
N° of + particle types :            0
N° of images :                      15

CHEMISTRY RESULT
pad                 Conv.           Arbitr.
BIL                 neg mg/dl       neg
UBG                 2 mg/dl         +
KET                 5 mg/dl         (+)
ASC                 20 mg/dl        +
GLU                 norm mg/dl      norm
PRO                 neg mg/dl       neg
BLD                 neg Ery/µl      neg
PH                  7.5             7.5
NIT                 pos             +
LEU                 25 Leu/µl       +
SG                  1.001
Turbidity                           Clear
Color                               Yellow

SEDIMENT RESULT
                     p/µl            N°                  +/-                 Ref.
(p/µl)
 RBC                0.0                                 -                   0 .. 10
 WBC                0.7                                 -                   0 .. 12
 CRY                0.0                                 -                   0 .. 6
 .CRY               0.0                                 -                   0 .. 6
 .CaOxm             0.0                                 -                   0 .. 6
```

```
 .CaOxd             0.0                                       -                  0 .. 6
 HYA                0.0                                       -                  0 .. 2
 PAT                0.0                                       -                  0 .. 1.5
 NEC                0.0                                       -                  0 .. 2
 EPI                0.0                                       -                  0 .. 5
 YEA                0.0                                       -                  0 .. 3
 BAC                12.8                                      -                  0 .. 130
 .BAC               0.0                                       -                  0 .. 130
 .BACr              2.6                                       -                  0 .. 130
 .BACc              10.1                                      -                  0 .. 130
 MUC                2.6                                       -                  0 .. 264

Comment : WBC neg and LEU pos
Dilution factor : 1.0
[ETX]
```

Note: in case of LabUMat2 result the PMC data are not transferred.

4.2.1.2 Chemical (AX-4280) result example:

```
[STX] URINE ANALYZIS REPORT


Laboratory :                        Please fill it in SETTINGS/Maintenance menu !

Sample ID :                         0121
Patient name :                      -
Date&Time :                         05/07/2021 12:22:22
General sAAediment result :           -
N° of + particle types :            0
N° of images :                      15


SEDIMENT RESULT
                    p/µl                N°                  +/-                 Ref.
(p/µl)
 RBC                0.0                                       -                  0 .. 10
 WBC                0.0                                       -                  0 .. 12
 CRY                0.9                                       -                  0 .. 6
 .CRY               0.9                                       -                  0 .. 6
 .CaOxm             0.0                                       -                  0 .. 6
 .CaOxd             0.0                                       -                  0 .. 6
 HYA                0.0                                       -                  0 .. 2
 PAT                0.0                                       -                  0 .. 1.5
 NEC                0.0                                       -                  0 .. 2
 EPI                0.0                                       -                  0 .. 5
 YEA                0.0                                       -                  0 .. 3
 BAC                47.1                                      -                  0 .. 130
 .BAC               0.0                                       -                  0 .. 130
 .BACr              4.8                                       -                  0 .. 130
 .BACc              42.2                                      -                  0 .. 130
 MUC                1.8                                       -                  0 .. 264
```

```
Comment :
Dilution factor : 1.0
[ETX]
```

4.2.1.3 Sediment QC result example:

```
 [STX] URINE ANALYZIS REPORT


Laboratory :                        Please fill it in SETTINGS/Maintenance menu !

Sample ID :                         0121
Patient name :                      QC_LOW
Date&Time :                         07/07/2021 10:05:40
General sediment result :           -
N° of + particle types :            0
N° of images :                      15


SEDIMENT RESULT
                     p/µl                N°                  +/-                 Ref.
(p/µl)
 RBC                0.0                                     -                   0 .. 10
 WBC                0.0                                     -                   0 .. 12
 CRY                0.0                                     -                   0 .. 6
 .CRY               0.0                                     -                   0 .. 6
 .CaOxm             0.0                                     -                   0 .. 6
 .CaOxd             0.0                                     -                   0 .. 6
 HYA                0.0                                     -                   0 .. 2
 PAT                0.0                                     -                   0 .. 1.5
 NEC                0.0                                     -                   0 .. 2
 EPI                0.0                                     -                   0 .. 5
 YEA                0.0                                     -                   0 .. 3
 BAC                11.4                                    -                   0 .. 130
 .BAC               0.0                                     -                   0 .. 130
 .BACr              0.4                                     -                   0 .. 130
 .BACc              11.0                                    -                   0 .. 130
 MUC                7.5                                     -                   0 .. 264

Comment : Test passed!
Dilution factor : 1.0
[ETX]
```

4.2.1.4 Chemical QC result example:

```
 [STX] URINE ANALYZIS REPORT


Laboratory :                        Please fill it in
SETTINGS/Maintenance menu !

Sample ID :                         0121
Patient name :                      QC Low
```

```
Date&Time :                             07/07/2021 10:04:26
General chemical result :               +
General sediment result :               invalid
N° of images :                          0

CHEMISTRY RESULT
pad                 Conv.               Arbitr.
BIL                 neg mg/dl           neg
UBG                 2 mg/dl             +
KET                 neg mg/dl           neg
ASC                 20 mg/dl            +
GLU                 norm mg/dl          norm
PRO                 neg mg/dl           neg
BLD                 neg Ery/µl          neg
PH                  7.5                 7.5
NIT                 pos                 +
LEU                 25 Leu/µl           +
SG                  1.001
Turbidity                               Clear
Color                                   Yellow

SEDIMENT RESULT - Invalid measurement

Comment :
Dilution factor : 1.0
[ETX]
```

4.2.1.5 Explanation

[CR]: Carriage return

[LF]: Line feed

[SPn]: number of space characters

[CHn]: The number of spaces to reach "n" character length including the previous field. Eg.: [CH20]: The number of spaces to reach 20 character length including the previous field.

[Labname]: name of the laboratory

[Sample ID]: the identifier code from the sample

[Patient name] : name of the patient

[Date&Time]: Date&time in the format what is set in the operating system

[General chemical result]: result of General chemical measurement

[General sediment result]: result of General sediment measurement

[N° of + particle types]: number of "+" particle types

[N° of images]: number of images

[PAD_NAME1]: abbreviation for the pads (chemical parameters)

[SI_Result]: result in SI

[CONV_Result]: result in conv.

[ARB_Result]: result in arbitr.

[PARTICLE_NAME1]: abbreviation for the sediment particles

[p/µl_Result]: quantity of the particles in p/µl

[N°_Result]: number of particles

[+/-_Result]: +/- Result

[Ref. (p/µl)_Result]: Ref. (p/µl)_Result

[Comment]: comment

[Dilution factor]: Dilution factor

5. Unidirectional communication with host computer - 'Unidirectional from 1.7'

5.1 Low Level Protocol between UriSed and host computer:

STX + [frame] + ETX + CRC + CR + LF

STX: Start of Text

ETX: End of Text

CRC: The CRC is a one byte (0-255) checksum character stored hexadecimal value in 2 characters. The checksum is computed by adding the binary values of the characters, keeping the least significant eight bits of the result. The checksum is initialized to zero with the <STX> character. Each characters in the message text is added to the checksum (modulo 256). The computation for the checksum does not include <STX> the checksum characters, or trailing <CR>and<LF>.

CR: Carriage return

LF: Line feed

5.2 Frame Types:

Code	Record name	Description
S2	Sediment summary results	Sediment summary result
CH2	Chemical summary result	Chemical summary result

5.3 Frame definitions:

5.3.1 UriSed Summary Sediment Results (S2)

5.3.1.1 Structure of the frame

```
S2|ANALYZER_ID|UID|BARCODE|BARCODE_FLAG|PATIENT_NAME|RACK|TUBE|OPERATOR|
DATE (YYYY/MM/DD)|HOUR(HH:MM:SS)|VALID_STATUS|REVIEW_STATUS|DILUTION FACTOR|
COMMENT|LEVEL_FLAG|MEAS_TYPE^QC_LEVEL|
VALIDATED^VALIDATED_BY^SEDIMENT_MEASURED_BY^SENT_TO_LIS_BY^EXPORTED_BY^PRINTED_BY|
SAMPLE_STATUS|RESERVED3|RESERVED4|RESERVED5|RESERVED6|RESERVED7|RESERVED8|
```

```
Test1^Result11^Units11^Result12^Units12^Result13^Units13^Result14^Units14^MOD_STATUS1|
Test1Res1|Test1Res2|Test1Res3|Test1Res4|
Test2^Result21^Units21^Result22^Units22^Result23^Units23^Result24^Units24^MOD_STATUS2|
Test2Res1|Test2Res2|Test2Res3|Test2Res4| … |
TestN^ResultN1^UnitsN1^ResultN2^UnitsN2^ResultN3^UnitsN3^ResultN4^UnitsN4^MOD_STATUSN|
TestNRes1|TestNRes2|TestNRes3|TestNRes4
```

5.3.1.2 Sediment result example:

```
[STX]S2|URM07000002|1|00210_70515104664|1|-
|2|10|12|2021/07/05|15:10:46|1|0|1.00||0|0^255|1^Automatic
validated^12^12^-^-|1|||||||RBC^-^^0.0^p/HPF^0.0^p/ul^^^0||||WBC^-
^^0.2^p/HPF^0.7^p/ul^^^0||||CRY^-^^0.1^p/HPF^0.4^p/ul^^^0|||||.CRY^-
^^0.1^p/HPF^0.4^p/ul^^^0|||||.CaOxm^-
^^0.0^p/HPF^0.0^p/ul^^^0|||||.CaOxd^-^^0.0^p/HPF^0.0^p/ul^^^0||||HYA^-
^^0.0^p/HPF^0.0^p/ul^^^0||||PAT^-^^0.0^p/HPF^0.0^p/ul^^^0||||NEC^-
^^0.0^p/HPF^0.0^p/ul^^^0||||EPI^-^^0.0^p/HPF^0.0^p/ul^^^0||||YEA^-
^^0.0^p/HPF^0.0^p/ul^^^0||||BAC^+^^63.6^p/HPF^279.8^p/ul^^^0|||||.BAC^-
^^0.0^p/HPF^0.0^p/ul^^^0|||||.BACr^-
^^7.0^p/HPF^30.8^p/ul^^^0|||||.BACc^+^^56.6^p/HPF^249.0^p/ul^^^0||||MUC^
-^^1.5^p/HPF^6.6^p/ul^^^0||||[ETX]D0[CR][LF]
```

5.3.1.3 Sediment QC result example:

```
[STX]S2|URS04100191|2|0121|0|QC_LOW|5|5|12|2021/07/07|10:05:40|1|0|1.00|T
est passed!|0|1^0|0^Not yet validated^Service^12^-^-
|0|||||||RBC^Passed^^0.0^p/HPF^0.0^p/ul^^^0||||WBC^Passed^^0.0^p/HPF^0.0
^p/ul^^^0||||[ETX]37[CR][LF]
```

5.3.1.4 Explanation

ANALYZER_ID: the factory number of the instrument (serial number)

UID: unique identification

- the sample’s unique identification
- in one database there are no 2 same UID-s

BARCODE_FLAG: (If the ID generate mode is sequence number, it is always 0.)

- 0 : barcode read
- 1 : barcode generated

RACK:

- the number of the rack

TUBE POSITION:

- the position of the tube

OPERATOR:

- the user’s name who sent the sample to LIS
    - independently from who measured the sample or who validated it (if it was)

VALID_STATUS:
- 0 : Invalid sample
- 1 : Valid sample

REVIEW_STATUS:
- 0 : Review is not necessary
- 1 : Review is necessary

LEVEL_FLAG:
- 0 : Normal sample level
- 1 : Low sample level

MEAS_TYPE:
- 0 : Normal measurement
- 1 : QC measurement

QC_LEVEL:
- -1 or 255 : In case of Normal measurement
  The value depends on the host decoding method.
- 0 : Low level QC
- 1 : High level QC

VALIDATED:
- 0 : not validated
- 1 : validated

SAMPLE_STATUS:
- 0 : Negative
- 1 : Positive
- 2 : Review
- 3 : Invalid (e.g. Skipped)
- 4 : Empty

RESERVED N: This part of the frame is reserved for future usage.

Structure of the Results:

TestN	category (particle) name
ResultN1	category result
UnitsN1	empty as it is a value without unit
ResultN2	value in p/HPF
UnitsN2	p/HPF
ResultN3	value in p/ul
UnitsN3	p/ul
ResultN4	N°
UnitsN4	empty as it is a value without unit
MOD_STATUSN	0: Not modified 1 : Modified

TestNResN: This part of the frame is reserved for future usage.

5.3.2 Chemical result (CH2)

5.3.2.1 Structure of the frame

```
CH2|ANALYZER_ID |UID|BARCODE|BARCODE_FLAG|RACK|TUBE|OPERATOR|DATE (YYYY/MM/DD)|
HOUR(HH:MM:SS)|LEVEL_FLAG|SEQUENCE_NUMBER|
VALIDATED^VALIDATED_BY^CHEMICAL_MEASURED_BY^SENT_TO_LIS_BY^EXPORTED_BY^PRINTED_BY|
SAMPLE_STATUS|PATIENT_NAME^COMMENT|MEAS_TYPE|SG^SG_VALID^TURB^TURB_VALID
^COL^COL_VALID|
RESERVED6|RESERVED7|RESERVED8|
Test1^Result11^Units11^Reserved11^Reserved12^Result12^Reserved13^MOD_STATUS1|Test1Res1|Te
st1Res2|Test1Res3| … |
TestN^ResultN1^UnitsN1^ReservedN1^ReservedN2^ResultN2^ReservedN3^MOD_STATUSN|TestNRes1
|TestNRes2|TestNRes3
```

5.3.2.2 AX-4280 chemical result example:

```
[STX]CH2||3|0121|0|1|9|12|2021/07/05|16:05:00|0|1|1^Automatic validated^-
^12^-^-|1|-^|0|-1.00^0^-1^0^-1^0||||GLU^50^mg/dl^^^+-^^0|||||PRO^^^^^-
^^0|||||BIL^^^^^-
^^0|||||URO^^^^^NORMAL^^0|||||PH^6.5^^^^^^0|||||BLD^^^^^-
^^0|||||KET^10^mg/dl^^^+1^^0|||||NIT^^^^^+2^^0|||||LEU^75^Leu/ul^^^^^0|||
||TURB^^^^^-
^^0|||||S.G.^1.000^^^^^^0|||||COLOR^COLORLESS^^^^^^0||||[ETX]B1[CR][LF]
```

Note: the following values can be visible in the transferred result at turbidity:

<table>
<tr><th>Result in user software</th><th>ResultN1</th><th>ResultN2</th></tr>
<tr><td>No parameter</td><td rowspan="4">(empty string)</td><td>(empty string)</td></tr>
<tr><td>Clear</td><td>-</td></tr>
<tr><td>Light Turbid</td><td>+1</td></tr>
<tr><td>Very Turbid</td><td>+2</td></tr>
</table>

5.3.2.3 LabUMat2 chemical result example in case of selected *SI* unit:

```
[STX]CH2|UPA8101003|5|0121|0|3|5|12|2021/07/07|09:45:39|0|0|0^Not yet
validated^Operator^12^-^-|1|-^WBC neg and LEU
pos|0|1.001^1^1^1^1^1||||BIL^neg^umol/l^^^neg^^0|||||UBG^35^umol/l^^^+^^0
|||||KET^0.5^mmol/l^^^(+)^^0|||||ASC^0.2^g/l^^^+^^0|||||GLU^norm^mmol/l^^
^norm^^0|||||PRO^neg^g/l^^^neg^^0|||||BLD^neg^Ery/ul^^^neg^^0|||||PH^7.5^
^^^7.5^^0|||||NIT^pos^^^^+^^0|||||LEU^25^Leu/ul^^^+^^0||||[ETX]E4[CR][LF]
```

5.3.2.4 Chemical (LabUMat2) QC result example in case of selected *Conv.* unit:

```
[STX]CH2|UPA8101003|9|0121|0|5|5|12|2021/07/07|10:04:26|0|0|1^Automatic
validated^Service^12^-^-|1|QC
Low^|1|1.001^2^1^1^2^1||||BIL^neg^mg/dl^^^neg^^0|||||UBG^2^mg/dl^^^+^^0|||||KET^
neg^mg/dl^^^neg^^0|||||ASC^20^mg/dl^^^+^^0|||||GLU^norm^mg/dl^^^norm^^0|||||PRO^
```

```
neg^mg/dl^^^neg^^0|||||BLD^neg^Ery/ul^^^neg^^0|||||PH^7.5^^^^7.5^^0|||||NIT^pos^
^^^+^^0|||||LEU^25^Leu/ul^^^+^^0||||[ETX]BD[CR]LF]
```

5.3.2.5 Explanation

ANALYZER_ID: the factory number of the instrument (serial number)

UID: unique identification

- the sample's unique identification
- in one database there are no 2 same UID-s

BARCODE_FLAG: (If the ID generate mode is sequence number, it is always 0.)

- 0 : barcode read
- 1 : barcode generated

LEVEL_FLAG:

- 0 : Normal sample level
- 1 : Low sample level

SEQUENCE NUMBER:

- in case of LabUMat2: 0
- in case of AutionMAX: the number is the value what AutionMAX sent

VALIDATED:

- 0 : not validated
- 1 : validated

SAMPLE_STATUS:

- 0 : Negative
- 1 : Positive
- 2 : Skip
- 3 : Invalid

MEAS_TYPE:

- 0 : Normal measurement
- 1 : QC measurement

SG:

- in case of AutionMAX: -1.00
- in case of LabUMat2: it gets the actual SG value (1.000-1.050)

SG_VALID:

- in case of AutionMAX: 0
- in case of LabUMat2: 1
    - exception: if the SG result became invalid, then: 0

TURB:

- in case of AutionMAX: -1
- in case of LabUMat2: it gets the turbidity value:

Result in user software	*TURB*
No parameter	*0*
Clear	*1*
Light Turbid	*2*
Very Turbid	*3*

TURB_VALID:

- in case of AutionMAX: 0
- in case of LabUMat2: 1

COL:

- in case of AutionMAX: -1
- in case of LabUMat2: it gets the color value:

Result in user software	*COL*
Other	*0*
Pale yellow	*1*
Yellow	*2*
Amber	*3*
Brown	*4*
Orange	*5*
Red	*6*
Green	*7*
No color parameter	*8*

COL_VALID:

- in case of AutionMAX: 0
- in case of LabUMat2: 1

RESERVED/Reserved N: This part of the frame is reserved for future usage.

Structure of the Results:

TestN	parameter name
ResultN1	result in Conv./SI unit
UnitsN1	unit (in Conv./SI)
Reserved1	reserved for future usage
Reserved2	reserved for future usage
ResultN2	result in Arbitr. unit
ReservedN3	reserved for future usage
MOD_STATUSN	0 : Not modified 1 : Modified

Abbreviations of the unit systems:

- *Conv.: conventional*
- *SI: Système international*

- *Arbitr.: arbitrary (a relative unit of measurement)*

Note1: the units depend on the actual parameters

Note2: in case of AutionMAX "TURB", "S.G." and "COLOR" are handled the same way as the other parameters

TestNResN: This part of the frame is reserved for future usage.

6. Bidirectional communication with host computer - 'Bidirectional up to 1.6.1'

6.1 Low Level Protocol between UriSed and host computer:

STX + [frame] + ETX + CRC + CR + LF

STX: Start of text (frame)

ETX: End of text (frame)

CRC: The CRC is a one byte (0-255) checksum character stored hexadecimal value in 2 characters. The checksum is computed by adding the binary values of the characters, keeping the least significant eight bits of the result. The checksum is initialized to zero with the <STX> character. Each characters in the message text is added to the checksum (modulo 256). The computation for the checksum does not include <STX> the checksum characters, or trailing <CR>and<LF>.

CR: Carriage return, recovery

LF: Line feed, start of new line

UriSed		**Host PC**
ASTM Frame	→	
	←	ACK / NAK
	←	ASTM Frame
ACK / NAK	→	

Only UriSed can open a communication. The communication will be initiated by sending E frame (see Frame definition). Host PC confirms its readiness by sending ACK (Hex 06) or signs a negative readiness by sending NAK (Hex 15). Under connection opening, there is "3 seconds timeout" between 2 attempts. After 3 NAK messages, host PC must break connection.

When the readiness is confirmed by host PC, the transmission will be sent frame by frame. The receiver has to confirm a frame reception with ACK. The receiver side can trigger a resending of the last sent frame with NAK (Max. 3 times for one frame).

If the ANALYZER_ID (in the sent message) doesn't match the ID from the Host, there will be sent NAK.

The communication phase will be ended by sending CL frame (see Frame definition).

6.2 Frame Types and High level protocol:

6.2.1 Frame Types

Code	Record name	Description
E	Enquiry	Connection open and check
W	Worklist	Download worklist from host PC
S	Sediment summary result	Sediment summary result
CH	Chemical summary result	Chemical summary result
QS2	Get sediment summary results with date interval (query sediment)	If the result is in date interval, and not sent to host PC. (Ex: error in connection)
CL	Close connection	For example: UriSed switch off.

6.2.2 High Level protocol between the UriSed and host PC:

- **UriSed** can send the following frames: E, S, CH, CL
- **Host PC** to UriSed can send the following frames: E, W, QS2, CL

6.3 Frame definitions:

6.3.1 Enquiry (E)

E|ANALYZER_ID

Example:
```
[STX]E|URM07000002[ETX]41[CR][LF]
```

6.3.2 Worklist (W)

6.3.2.1 Structure of the frame

W|ANALYZER_ID|Barcode1^PatientName1^Comment1^Chemical1^Sediment1|Barcode2^PatientName 2^Comment2^ Chemical2^Sediment2| ... | BarcodeN^PatientNameN^CommentN^ ChemicalN^SedimentN

Chemical flag (applied to chemical result, if LabUMat2 is connected):
- 0: measurement is not needed on the instrument
- 1: measurement is needed on the instrument

Sediment flag (applied to sediment result):
- 0: measurement is not needed on the instrument
- 1: measurement is needed on the instrument

6.3.2.2 Example:

```
[STX]
W|URM07000002|1111^John^Comment^0^0|2222^Noel^Comm^0^1|333^Eva^Comm^1^0
[ETX]9A[CR][LF]
```

6.3.3 UriSed Summary Sediment Results (S)

6.3.3.1 Structure of the frame

S|ANALYZER_ID|UID|BARCODE|BARCODE_FLAG|PATIENT_NAME|RACK|TUBE|OPERATOR| DATE (YYYY/MM/DD)|HOUR(HH:MM:SS)|VALID_STATUS|REVIEW_STATUS|DILUTION FACTOR| COMMENT| Test1^Result11^Units11^Result12^Units12^Result13^Units13^Result14^Units14^MOD_STATUS1| Test2^Result21^Units21^Result22^Units22^Result23^Units23^Result24^Units24^MOD_STATUS2| … | TestN^ResultN1^UnitsN1^ResultN2^UnitsN2^ResultN3^UnitsN3^ResultN4^UnitsN4^MOD_STATUSN

6.3.3.2 Sediment result example:

```
[STX]S|URM07000002|10|00306_709455045|1|-
|3|6|12|2021/07/07|09:45:50|0|0|1.00||RBC^N/A^^^p/HPF^^p/ul^^^0|WBC^N/A^^^p/HPF^
^p/ul^^^0|CRY^N/A^^^p/HPF^^p/ul^^^0|.CRY^N/A^^^p/HPF^^p/ul^^^0|.CaOxm^N/A^^^p/HP
F^^p/ul^^^0|.CaOxd^N/A^^^p/HPF^^p/ul^^^0|HYA^N/A^^^p/HPF^^p/ul^^^0|PAT^N/A^^^p/H
PF^^p/ul^^^0|NEC^N/A^^^p/HPF^^p/ul^^^0|EPI^N/A^^^p/HPF^^p/ul^^^0|YEA^N/A^^^p/HPF
^^p/ul^^^0|BAC^N/A^^^p/HPF^^p/ul^^^0|.BAC^N/A^^^p/HPF^^p/ul^^^0|.BACr^N/A^^^p/HP
F^^p/ul^^^0|.BACc^N/A^^^p/HPF^^p/ul^^^0|MUC^N/A^^^p/HPF^^p/ul^^^0[ETX]D6[CR][LF]
```

6.3.3.3 Sediment QC result example:

```
[STX]S|URS04100191|12|0121|0|QC_LOW|5|5|12|2021/07/07|10:05:40|1|0|1.00|Test
passed!|RBC^Passed^^0.0^p/HPF^0.0^p/ul^^^0|WBC^Passed^^0.0^p/HPF^0.0^p/ul^^^0[ET
X]5A[CR][LF]
```

6.3.3.4 Explanation

ANALYZER_ID: the factory number of the instrument (serial number)

UID: unique identification

- the sample's unique identification
- in one database there are no 2 same UID-s

BARCODE_FLAG: (If the ID generate mode is sequence number, it is always 0.)

- 0 : barcode read
- 1 : barcode generated

VALID_STATUS:

- 0 : Invalid sample
- 1 : Valid sample

REVIEW_STATUS:

- 0 : Review is not necessary
- 1 : Review is necessary

Structure of the Results:

TestN	category (particle) name
ResultN1	category result
UnitsN1	empty as it is a value without unit
ResultN2	value in p/HPF
UnitsN2	p/HPF
ResultN3	value in p/ul
UnitsN3	p/ul
ResultN4	N°
UnitsN4	empty as it is a value without unit
MOD_STATUSN	0: Not modified 1 : Modified

6.3.4 Chemical result (CH)

Chemical result frame (CH) always sends with Sediment result frame. If the result in the database, don't contains Sediment results, (it has just chemical result,) then the sediment frame values will be empty. It is true for the QC case, too.

6.3.4.1 Structure of the frame

CH|ANALYZER_ID |UID|BARCODE|BARCODE_FLAG|RACK|TUBE|OPERATOR|DATE (YYYY/MM/DD)| HOUR(HH:MM:SS)| Test1^Result11^Units11^Reserved11^Reserved12^Result12^ReservedN13^MOD_STATUS1|… | TestN^ResultN1^UnitsN1^ReservedN1^ReservedN2^ResultN2^RESERVEDN2^MOD_STATUSN

6.3.4.2 AX-4280 chemical result example:

```
[STX]CH||15|0121|0|1|9|12|2021/07/05|16:05:00|GLU^50^mg/dl^^^+-^^0|PRO^^^^^-
^^0|BIL^^^^^-^^0|URO^^^^^NORMAL^^0|PH^6.5^^^^^^0|BLD^^^^^-
^^0|KET^10^mg/dl^^^+1^^0|NIT^^^^^+2^^0|LEU^75^Leu/ul^^^^^0|TURB^^^^^-
^^0|S.G.^1.000^^^^^^0|COLOR^COLORLESS^^^^^^0[ETX]D9[CR][LF]
```

The following values can be visible in the transferred result at turbidity:

Result in user software	*ResultN1*	*ResultN2*
No parameter	*(empty string)*	*(empty string)*
Clear		*-*
Light Turbid		*+1*
Very Turbid		*+2*

6.3.4.3 LabUMat2 chemical result example in case of selected *SI* unit:

```
[STX]CH|UPA8101003|17|00306_709455045|1|3|6|12|2021/07/07|09:45:50|BIL^neg^umol/
l^^^neg^^0|UBG^35^umol/l^^^+^^0|KET^0.5^mmol/l^^^(+)^^0|ASC^0.2^g/l^^^+^^0|GLU^n
orm^mmol/l^^^norm^^0|PRO^neg^g/l^^^neg^^0|BLD^neg^Ery/ul^^^neg^^0|PH^7.5^^^^7.5^
^0|NIT^pos^^^^+^^0|LEU^25^Leu/ul^^^+^^0[ETX]8B[CR][LF]
```

Note: the PMC parameters (SG, Turbidity, Color) are not visible

6.3.4.4 Chemical (LabUMat2) QC result example in case of selected *Conv.* unit:

```
[STX]CH|UPA8101003|19|0121|0|5|5|12|2021/07/07|10:04:26|BIL^neg^mg/dl^^^neg^^0|U
BG^2^mg/dl^^^+^^0|KET^neg^mg/dl^^^neg^^0|ASC^20^mg/dl^^^+^^0|GLU^norm^mg/dl^^^no
rm^^0|PRO^neg^mg/dl^^^neg^^0|BLD^neg^Ery/ul^^^neg^^0|PH^7.5^^^^7.5^^0|NIT^pos^^^
^+^^0|LEU^25^Leu/ul^^^+^^0[ETX]B7[CR][LF]
```

Note: the PMC parameters (SG, Turbidity, Color) are not visible

6.3.4.5 Explanation

ANALYZER_ID: the factory number of the instrument (serial number)

UID: unique identification

- the sample's unique identification

- in one database there are no 2 same UID-s

BARCODE_FLAG: (If the ID generate mode is sequence number, it is always 0.)
- 0 : barcode read
- 1 : barcode generated

Reserved N: This part of the frame is reserved for future usage.

Structure of the Results:

TestN	parameter name
ResultN1	result in Conv./SI unit
UnitsN1	unit (in Conv./SI)
ReservedN1	reserved for future usage
ReservedN2	reserved for future usage
ResultN2	result in Arbitr. unit
ReservedN3	reserved for future usage
MOD_STATUSN	0 : Not modified 1 : Modified

Note1: the units depend on the measurement settings
Note2: in case of AutionMAX "TURB", "S.G." and "COLOR" are handled the same way as the other parameters

6.3.5 UriSed Summary Sediment Results with date interval query (QS2)

QS2|ANALYZER_ID|FROM DATE(YYYY/MM/DD)|TO DATE(YYYY/MM/DD)

Example:
`[STX]QS2|URM07000002|2021/07/05|2021/07/06[ETX]A9[CR][LF]`

6.3.6 Close connection (CL)

CL|ANALYZER_ID

Example:
`[STX]CL|URM07000002[ETX]8B[CR][LF]`

7. Bidirectional communication with host computer - 'Bidirectional from 1.7'

7.1 Low Level Protocol between UriSed and host computer:

STX + [frame] + ETX + CRC + CR + LF

STX: Start of text (frame)

ETX: End of text (frame)

CRC: The CRC is a one byte (0-255) checksum character stored hexadecimal value in 2 characters. The checksum is computed by adding the binary values of the characters, keeping the least significant eight bits of the result. The checksum is initialized to zero with the <STX> character. Each characters in the message text is added to the checksum (modulo 256). The computation for the checksum does not include <STX> the checksum characters, or trailing <CR>and<LF>.

CR: Carriage return, recovery

LF: Line feed, start of new line

UriSed		**Host PC**
ASTM Frame	→	
	←	ACK / NAK
	←	ASTM Frame
ACK / NAK	→	

Only UriSed can open a communication. The communication will be initiated by sending E frame (see Frame definition). Host PC confirms its readiness by sending ACK (Hex 06) or signs a negative readiness by sending NAK (Hex 15). Under connection opening, there is “3 seconds timeout” between 2 attempts. After 3 NAK messages, host PC must break connection.

When the readiness is confirmed by host PC, the transmission will be sent frame by frame. The receiver has to confirm a frame reception with ACK. The receiver side can trigger a resending of the last sent frame with NAK (Max. 3 times for one frame).

If the ANALYZER_ID (in the sent message) doesn’t match the ID from the Host, there will be sent NAK.

The communication phase will be ended by sending CL frame (see Frame definition).

7.2 Frame Types and High level protocol:

7.2.1 Frame Types

Code	Record name	Description
E	Enquiry	Connection open and check
W	Worklist	Download worklist from host PC
S2	Sediment summary result	Sediment summary result
CH2	Chemical summary result	Chemical summary result
QS2	Get sediment summary results with date interval (query sediment)	If the result is in date interval, and not sent to host PC. (Ex: error in connection)
CL	Close connection	For example UriSed switch off.

7.2.2 High Level protocol between the UriSed and host PC:

- **UriSed** can send the following frames: E, S2, CH2, CL
- **Host PC** can send to UriSed the following frames: E, W, QS2, CL

7.3 Frame definitions:

7.3.1 Enquiry (E)

E|ANALYZER_ID

Example:
```
[STX]E|URM07000002[ETX]41[CR][LF]
```

7.3.2 Worklist (W)

7.3.2.1 Structure of the frame

W|ANALYZER_ID|Barcode1^PatientName1^Comment1^Chemical1^Sediment1|Barcode2^PatientName2^Comment2^ Chemical2^Sediment2| ... | BarcodeN^PatientNameN^CommentN^ ChemicalN^SedimentN

Chemical flag (applied to chemical result, if LabUMat2 is connected):
- 0: measurement is not needed on the instrument
- 1: measurement is needed on the instrument

Sediment flag (applied to sediment result):
- 0: measurement is not needed on the instrument
- 1: measurement is needed on the instrument

7.3.2.2 Example

```
[STX]W|URM07000002|1111^John^Comment^0^0|2222^Noel^Comm^0^1|333^Eva^Comm^1^0[ETX
]9A[CR][LF]
```

7.3.3 UriSed Summary Sediment Results (S2)

7.3.3.1 Structure of the frame

S2|ANALYZER_ID |UID|BARCODE|BARCODE_FLAG|PATIENT_NAME|RACK|TUBE|OPERATOR| DATE (YYYY/MM/DD)|HOUR(HH:MM:SS)|VALID_STATUS|REVIEW_STATUS|DILUTION FACTOR| COMMENT|LOW_LEVEL|MEAS_TYPE^QC_LEVEL| VALIDATED^VALIDATED_BY^SEDIMENT_MEASURED_BY^SENT_TO_LIS_BY^EXPORTED_BY^PRINTED_BY| SAMPLE_STATUS|RESERVED3|RESERVED4|RESERVED5|RESERVED6|RESERVED7|RESERVED8| Test1^Result11^Units11^Result12^Units12^Result13^Units13^Result14^Units14^MOD_STATUS1| Test1Res1|Test1Res2|Test1Res3|Test1Res4| Test2^Result21^Units21^Result22^Units22^Result23^Units23^Result24^Units24^MOD_STATUS2| Test2Res1|Test2Res2|Test2Res3|Test2Res4| … | TestN^ResultN1^UnitsN1^ResultN2^UnitsN2^ResultN3^UnitsN3^ResultN4^UnitsN4^MOD_STATUSN| TestNRes1|TestNRes2|TestNRes3|TestNRes4

7.3.3.2 Sediment result example :

```
[STX]S2|URM07000002|1|0121|0|-
|2|9|SuperUser|2021/07/05|15:10:30|1|0|1.00||0|0^255|1^Automatic
validated^12^SuperUser^-^-|0|||||||RBC^-^^0.2^p/HPF^0.9^p/ul^^^0|||||WBC^-
^^0.0^p/HPF^0.0^p/ul^^^0|||||CRY^-^^0.1^p/HPF^0.5^p/ul^^^0|||||.CRY^-
^^0.1^p/HPF^0.5^p/ul^^^0|||||.CaOxm^-^^0.0^p/HPF^0.0^p/ul^^^0|||||.CaOxd^-
^^0.0^p/HPF^0.0^p/ul^^^0|||||HYA^-^^0.0^p/HPF^0.0^p/ul^^^0|||||PAT^-
^^0.0^p/HPF^0.0^p/ul^^^0|||||NEC^-^^0.0^p/HPF^0.0^p/ul^^^0|||||EPI^-
^^0.0^p/HPF^0.0^p/ul^^^0|||||YEA^-^^0.0^p/HPF^0.0^p/ul^^^0|||||BAC^-
^^7.6^p/HPF^33.5^p/ul^^^0|||||.BAC^-^^0.0^p/HPF^0.0^p/ul^^^0|||||.BACr^-
^^0.4^p/HPF^1.9^p/ul^^^0|||||.BACc^-^^7.2^p/HPF^31.6^p/ul^^^0|||||MUC^-
^^0.3^p/HPF^1.4^p/ul^^^0|||||[ETX]5C[CR][LF]
```

7.3.3.3 Sediment QC result example:

```
[STX]S2|URS04100191|2|0121|0|QC_LOW|5|5|SuperUser|2021/07/07|10:05:40|1|0|1.00|T
est passed!|0|1^0|0^Not yet validated^Service^SuperUser^-^-
|0|||||||RBC^Passed^^0.0^p/HPF^0.0^p/ul^^^0|||||WBC^Passed^^0.0^p/HPF^0.0^p/ul^^
^0|||||[ETX]CD[CR][LF]
```

7.3.3.4 Explanation

ANALYZER_ID: the factory number of the instrument (serial number)

UID: unique identification

- the sample's unique identification
- in one database there are no 2 same UID-s

BARCODE_FLAG: (If the ID generate mode is sequence number, it is always 0.)

- 0 : barcode read
- 1 : barcode generated

VALID_STATUS:

- 0 : Invalid sample
- 1 : Valid sample

REVIEW_STATUS:

- 0 : Review is not necessary
- 1 : Review is necessary

LOW_LEVEL:

- 0 : Normal sample level
- 1 : Low sample level

MEAS_TYPE:

- 0 : Normal measurement
- 1 : QC measurement

QC_LEVEL:

- -1 or 255 : In case of Normal measurement
  The value depends on the host decoding method.
- 0 : Low level QC

- 1 : High level QC

VALIDATED:

- 0 : not validated
- 1 : validated

SAMPLE_STATUS:

- 0 : Negative
- 1 : Positive
- 2 : Review
- 3 : Invalid (e.g. Skipped)
- 4 : Empty

RESERVED N: This part of the frame is reserved for future usage.

Structure of the Results:

TestN	category (particle) name
ResultN1	category result
UnitsN1	empty as it is a value without unit
ResultN2	value in p/HPF
UnitsN2	p/HPF
ResultN3	value in p/ul
UnitsN3	p/ul
ResultN4	N°
UnitsN4	empty as it is a value without unit
MOD_STATUSN	0: Not modified 1 : Modified

TestNResN: This part of the frame is reserved for future usage.

7.3.4 Chemical result (CH2)

7.3.4.1 Structure of the frame

```
CH2|ANALYZER_ID |UID|BARCODE|BARCODE_FLAG|RACK|TUBE|OPERATOR|DATE (YYYY/MM/DD)|
HOUR(HH:MM:SS)|LEVEL_FLAG|SEQUENCE_NUMBER|
VALIDATED^VALIDATED_BY^CHEMICAL_MEASURED_BY^SENT_TO_LIS_BY^EXPORTED_BY^PRINTED_BY|
SAMPLE_STATUS|PATIENT_NAME^COMMENT|MEAS_TYPE|SG^SG_VALID^TURB^TURB_VALID^COL^CO
L_VALID|
RESERVED6|RESERVED7|RESERVED8|
Test1^Result11^Units11^Reserved11^Reserved12^Result12^Reserved13^MOD_STATUS1|Test1Res1|Te
st1Res2|Test1Res3| … |
TestN^ResultN1^UnitsN1^ReservedN1^ReservedN1^ResultN2^ReservedN3^MOD_STATUSN|TestNRes1
|TestNRes2|TestNRes3
```

7.3.4.2 AX-4280 chemical result example:

```
[STX] CH2||3|0121|0|1|9|SuperUser|2021/07/05|16:05:00|0|1|1^Automatic
validated^-^SuperUser^-^-|1|-^|0|-1.00^0^-1^0^-1^0||||GLU^50^mg/dl^^^+-
^^0|||||PRO^^^^^-^^0|||||BIL^^^^^-
^^0|||||URO^^^^^NORMAL^^0|||||PH^6.5^^^^^^0|||||BLD^^^^^-
^^0|||||KET^10^mg/dl^^^+1^^0|||||NIT^^^^^+2^^0|||||LEU^75^Leu/ul^^^^^0|||||TURB^
^^^^-^^0|||||S.G.^1.000^^^^^^0|||||COLOR^COLORLESS^^^^^^0||||[ETX]47[CR][LF]
```

Note: the following values can be visible in the transferred result at turbidity:

Result in user software	*ResultN1*	*ResultN2*
No parameter	*(empty string)*	*(empty string)*
Clear		*-*
Light Turbid		*+1*
Very Turbid		*+2*

7.3.4.3 LabUMat2 chemical result example in case of selected *Conv.* unit:

```
[STX]CH2|UPA8101003|4|00406_709494115|1|4|6|SuperUser|2021/07/07|09:49:41|0|0|1^
Automatic validated^Service^SuperUser^-^-|1|-
^|0|1.001^1^1^1^0^1||||BIL^neg^mg/dl^^^neg^^0|||||UBG^2^mg/dl^^^+^^0|||||KET^5^m
g/dl^^^(+)^^0|||||ASC^20^mg/dl^^^+^^0|||||GLU^norm^mg/dl^^^norm^^0|||||PRO^neg^m
g/dl^^^neg^^0|||||BLD^neg^Ery/ul^^^neg^^0|||||PH^7.5^^^^7.5^^0|||||NIT^pos^^^^+^
^0|||||LEU^25^Leu/ul^^^+^^0||||[ETX]4A[CR][LF]
```

7.3.4.4 Chemical QC result example in case of selected *SI* unit:

```
[STX]CH2|UPA8101003|6|00606_710124993|1|6|6|SuperUser|2021/07/07|10:12:50|0|0|1^
Automatic validated^Service^SuperUser^-^-|1|QC
High^|1|1.001^2^1^1^1^1||||BIL^neg^umol/l^^^neg^^0|||||UBG^35^umol/l^^^+^^0|||||
KET^neg^mmol/l^^^neg^^0|||||ASC^0.2^g/l^^^+^^0|||||GLU^norm^mmol/l^^^norm^^0||||
|PRO^neg^g/l^^^neg^^0|||||BLD^neg^Ery/ul^^^neg^^0|||||PH^7.5^^^^7.5^^0|||||NIT^p
os^^^^+^^0|||||LEU^25^Leu/ul^^^+^^0||||[ETX]CD[CR][LF]
```

7.3.4.5 Explanation

ANALYZER_ID: the factory number of the instrument (serial number)

UID: unique identification

- the sample's unique identification
- in one database there are no 2 same UID-s

BARCODE_FLAG: (If the ID generate mode is sequence number, it is always 0.)

- 0 : barcode read
- 1 : barcode generated

LEVEL_FLAG:

- 0 : Normal sample level
- 1 : Low sample level

SEQUENCE NUMBER:

- in case of LabUMat2: 0

- in case of AutionMAX: the number is the value what AutionMAX sent

VALIDATED:

- 0 : not validated
- 1 : validated

SAMPLE_STATUS:

- 0 : Negative
- 1 : Positive
- 2 : Skip
- 3 : Invalid

MEAS_TYPE:

- 0 : Normal measurement
- 1 : QC measurement

SG:

- in case of AutionMAX: -1.00
- in case of LabUMat2: it gets the actual SG value (1.000-1.050)

SG_VALID:

- 0: Invalid
- 1: Normal (passed)
- 2: Abnormal (failed)

TURB:

- in case of AutionMAX: -1
- in case of LabUMat2: it gets the turbidity value:

Result in user software	*TURB*
No parameter	*0*
Clear	*1*
Light Turbid	*2*
Very Turbid	*3*

TURB_VALID:

- in case of AutionMAX: 0
- in case of LabUMat2: 1

COL:

- in case of AutionMAX: -1
- in case of LabUMat2: it gets the color value:

Result in user software	*COL*
Other	*0*
Pale yellow	*1*
Yellow	*2*
Amber	*3*
Brown	*4*
Orange	*5*
Red	*6*
Green	*7*
No color parameter	*8*

COL_VALID:

- in case of AutionMAX: 0
- in case of LabUMat2: 1

RESERVED/Reserved N: This part of the frame is reserved for future usage.

Structure of the Results:

TestN	parameter name
ResultN1	result in Conv./SI unit
UnitsN1	unit (in Conv./SI)
ReservedN1	reserved for future usage
ReservedN2	reserved for future usage
ResultN2	result in Arbitr. unit
ReservedN3	reserved for future usage
MOD_STATUSN	0 : Not modified 1 : Modified

Abbreviations of the unit systems:

- *Conv.: conventional*
- *SI: Système international*
- *Arbitr.: arbitrary (a relative unit of measurement)*

Note1: the units depend on the actual parameters

Note2: in case of AutionMAX "TURB", "S.G." and "COLOR" are handled the same way as the other parameters

TestNResN: This part of the frame is reserved for future usage.

7.3.5 UriSed Summary Sediment Results with date interval query (QS2)

QS2|ANALYZER_ID|FROM DATE(YYYY/MM/DD)|TO DATE(YYYY/MM/DD)

Example:
`[STX]QS2|URM07000002|2021/07/05|2021/07/06[ETX]A9[CR][LF]`

7.3.6 Close connection (CL)

CL|ANALYZER_ID

Example:
`[STX]CL|URM07000002[ETX]8B[CR][LF]`

8. LIS2 A2 Low level interface

8.1 Frame structure:

[STX]	*{FN}*	Data content of Message	[ETX]	*{CHS}*	[CR]	[LF]

The legend of the colouring of the fields of the high level protocol is the following:

(purple)	Fixed fields
(green)	Common fields
	Data field

8.1.1 Fixed fields

The LIS standard has constraints for some fields. These are marked by the legend as fixed fields. They cannot be changed, and will be the same static content over the entire communication in all frames.

8.1.2 Common fields

These fields content changes during the communication but the content is generated by the protocol according to rules which are applied for all frames.

8.1.3 Data content

Data content varies for each type of records. The details of each records can be found in this document later.

Abbreviation	**FN**
Short explanation	Frame Number
Description	A single digit from 0 to 7 The frame number begins at 1 with the first frame of the transfer phase. The frame number is incremented by one for every new frame transmitted. After 7, the frame number rolls over to 0 and continuous in this fashion.
Example	1; 2

Abbreviation	**CHS**
Short explanation	Checksum
Description	The checksum permits the receiver to detect a defective frame. The checksum is encoded as two characters. Which are sent after the <ETB> or <ETX> characters. The checksum is computed by adding the binary values of the characters, keeping the least significant eight bits of the result. • The checksum is initialized to zero with the <STX> character. The first character used in computing the checksum is the frame number. Each character in the message text, and the <ETB>/<ETX> are added to the checksum (modulo 256). The computation for the checksum does not include <STX> the checksum characters, or trailing <CR>and<LF>. • The checksum is an integer represented by eight bits, it can be considered as two groups of four bits. The groups of four bits are converted to the ASCII characters of the hexadecimal representation.
Example	DE; 94; D6

9. LIS2 A2 High level interface

The base of the high level layer of the data exchange is described in the LIS2 A2 standard.

9.1 Characters and encoding

The UriSed sends and receives strings with ASCII character encoding.

9.1.1 Delimiters

For the purpose of providing examples, the following delimiters are used in this description:

Delimiter type	Name of the character	Sign of the character	ASCII code of the character
Field delimiter	vertical bar	|	124
Repeat delimiter	backslash	\	92
Component delimiter	caret	^	94
Escape delimiter	ampersand	&	38

9.2 Message Types

The following message types exist in LIS2-A2 implementation of the UriSed instrument:

- Measurement messages: (initiated only by the UriSed)
  - Sample Measurement Message
  - QC Measurement Message
  - Host Query Message

- Request messages (sent from host to UriSed):
  - Result Query Message (load list)
  - Worklist Entry Message
  - Host Query Response Message
  - Sending Images

9.2.1 Sample Measurement Message

The Sample Measurement Message contains the data for a single measurement by the UriSed instrument. The Sample Measurement Message sending is always initiated by the UriSed.

In case of a connected result the sediment and the chemical results are sent one after another, but the result structures are the same as in case of a standalone result. The result sending order can be set in the user software.

General structure:

- Header[1]
    - Patient[2] #1 (general information about the first patient)
        - Comment[3] #1 (optional; related to patient #1)
        - Order[4] #1(information about the first test order; it related to patient #1)
            - Comment[5] #1 (optional; related to order #1)
            - Result[6] #1 (result information related to order #1)
                - Comment #1(optional; related to result #1)
                - Comment #2(optional; related to result #1)

              ...
            - Result #2 (result information related to order #1)

              ...
            - Result #n (result information related to order #1)
        - Order #2 (information about the second test order; it relates to patient #1)
            - Result #1 (result information related to order #2)
            - Result #2 (result information related to order #2)

          ...
        - Order #n (information about the last test order; it relates to patient #1)
            - Result #1 (result information relates to order #n)

          ...
    - Patient #2 (all of the structures repeats)

      ...
    - Patient #n
- Message Terminator[7]

Structure implementation on UriSed:

- Message Header

[1] see also Message Header Record (H)

[2] see also

Patient Identifying Record (P)

[3] see also Comment Record (C)

[4] see also Test Order Record (O)

[5] see also Comment Record (C)

[6] see also Result Record (R)

[7] see also Message Terminator Record (L)

        - Patient (general information about the patient)
        - Comment (general comment for sample)
            - Order (information about the test order)
                - Result #1 (result related to order #1)
                - Comment #1 (related to result #1)
                - Result #2 (result related to order #1)
                - Comment #2 (related to result #2)
                - ......
                - Result #n (result information relates to order #1)
                - Comment #n (related to result #n)
- Message Terminator

Result (R) **record units**: UriSed sends the sediment results for every parameter in each display unit:

- p/HPF
- p/LPF
- p/ul
- Number
- Category

In case of chemical result UriSed sends the result in these units: on chemical analyzer selected unit + the arbitrary value. (It depends on the chemical analyzer type.)

The examples show the sequence of the communications. The **BLACK** color is the Host and the **GREEN** color is the system (UriSed 3 PRO).

9.2.1.1 Example 1: Sediment Sample Measurement Message transfer

```
[ENQ]
[ACK]
[STX]1H|\^&|||UriSed 3 PRO^UriSed 3 PRO^4.3.45.8513^1^URM07000002||||||P|LIS2-
A2|20210706085625[CR][ETB]50[CR][LF]
[ACK]
[STX]2P|1||||-[CR][ETB]70[CR][LF]
[ACK]
[STX]3O|1|00210_70515104664^G|2^10^12^SAMPLE|S^^^Cuvette^^|R|||||N|||20210705151046||||
||||||F[CR][ETB]A4[CR][LF]
[ACK]
[STX]4R|1|46419-8^^^RBC|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]37[CR][LF]
[ACK]
[STX]5R|2|798-9^^^RBC|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]DD[CR][LF]
[ACK]
[STX]6R|3|53292-9^^^RBC|-|||N||F||12|||UriSed 3 PRO[CR][ETB]5B[CR][LF]
[ACK]
[STX]7R|4|46702-7^^^WBC|0.2|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]3E[CR][LF]
[ACK]
[STX]0R|5|51487-7^^^WBC|0.7|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]46[CR][LF]
[ACK]
[STX]1R|6|53316-6^^^WBC|-|||N||F||12|||UriSed 3 PRO[CR][ETB]58[CR][LF]
[ACK]
```

```
[STX]2R|7|53307-5^^^CRY|0.1|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]4A[CR][LF]
[ACK]
[STX]3R|8|53297-8^^^CRY|0.4|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]5D[CR][LF]
[ACK]
[STX]4R|9|53334-9^^^CRY|-|||N||F||12|||UriSed 3 PRO[CR][ETB]73[CR][LF]
[ACK]
[STX]5R|10|53307-5^^^.CRY|0.1|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]A5[CR][LF]
[ACK]
[STX]6R|11|53297-8^^^.CRY|0.4|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]B8[CR][LF]
[ACK]
[STX]7R|12|53334-9^^^.CRY|-|||N||F||12|||UriSed 3 PRO[CR][ETB]CE[CR][LF]
[ACK]
[STX]0R|13|^^^.CaOxm|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]28[CR][LF]
[ACK]
[STX]1R|14|^^^.CaOxm|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]2D[CR][LF]
[ACK]
[STX]2R|15|^^^.CaOxm|-|||N||F||12|||UriSed 3 PRO[CR][ETB]4E[CR][LF]
[ACK]
[STX]3R|16|^^^.CaOxd|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]25[CR][LF]
[ACK]
[STX]4R|17|^^^.CaOxd|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]2A[CR][LF]
[ACK]
[STX]5R|18|38993-2^^^.CaOxd|-|||N||F||12|||UriSed 3 PRO[CR][ETB]BA[CR][LF]
[ACK]
[STX]6R|19|33223-9^^^HYA|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]73[CR][LF]
[ACK]
[STX]7R|20|51484-4^^^HYA|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]73[CR][LF]
[ACK]
[STX]0R|21|50231-0^^^HYA|-|||N||F||12|||UriSed 3 PRO[CR][ETB]7D[CR][LF]
[ACK]
[STX]1R|22|^^^PAT|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]08[CR][LF]
[ACK]
[STX]2R|23|^^^PAT|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]0D[CR][LF]
[ACK]
[STX]3R|24|72224-9^^^PAT|-|||N||F||12|||UriSed 3 PRO[CR][ETB]95[CR][LF]
[ACK]
[STX]4R|25|53294-5^^^NEC|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]68[CR][LF]
[ACK]
[STX]5R|26|51485-1^^^NEC|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]69[CR][LF]
[ACK]
[STX]6R|27|50225-2^^^NEC|-|||N||F||12|||UriSed 3 PRO[CR][ETB]82[CR][LF]
[ACK]
[STX]7R|28|33219-7^^^EPI|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]73[CR][LF]
[ACK]
[STX]0R|29|51486-9^^^EPI|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]78[CR][LF]
[ACK]
[STX]1R|30|53318-2^^^EPI|-|||N||F||12|||UriSed 3 PRO[CR][ETB]85[CR][LF]
[ACK]
[STX]2R|31|^^^YEA|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]03[CR][LF]
[ACK]
[STX]3R|32|51481-0^^^YEA|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]68[CR][LF]
[ACK]
[STX]4R|33|72223-1^^^YEA|-|||N||F||12|||UriSed 3 PRO[CR][ETB]87[CR][LF]
[ACK]
```

```
[STX]5R|34|33218-9^^^BAC|63.6|p/HPF||A||F||12|||UriSed 3 PRO[CR][ETB]89[CR][LF]
[ACK]
[STX]6C|1|I|A|I[CR][ETB]91[CR][LF]
[ACK]
[STX]7R|35|51480-2^^^BAC|279.8|p/ul||A||F||12|||UriSed 3 PRO[CR][ETB]C4[CR][LF]
[ACK]
[STX]0C|1|I|A|I[CR][ETB]8B[CR][LF]
[ACK]
[STX]1R|36|50221-1^^^BAC|+|||A||F||12|||UriSed 3 PRO[CR][ETB]59[CR][LF]
[ACK]
[STX]2C|1|I|A|I[CR][ETB]8D[CR][LF]
[ACK]
[STX]3R|37|33218-9^^^.BAC|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]86[CR][LF]
[ACK]
[STX]4R|38|51480-2^^^.BAC|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]85[CR][LF]
[ACK]
[STX]5R|39|50221-1^^^.BAC|-|||N||F||12|||UriSed 3 PRO[CR][ETB]9D[CR][LF]
[ACK]
[STX]6R|40|^^^.BACr|7.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]95[CR][LF]
[ACK]
[STX]7R|41|^^^.BACr|30.8|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]CE[CR][LF]
[ACK]
[STX]0R|42|^^^.BACr|-|||N||F||12|||UriSed 3 PRO[CR][ETB]AC[CR][LF]
[ACK]
[STX]1R|43|^^^.BACc|56.6|p/HPF||A||F||12|||UriSed 3 PRO[CR][ETB]B1[CR][LF]
[ACK]
[STX]2C|1|I|A|I[CR][ETB]8D[CR][LF]
[ACK]
[STX]3R|44|^^^.BACc|249.0|p/ul||A||F||12|||UriSed 3 PRO[CR][ETB]E5[CR][LF]
[ACK]
[STX]4C|1|I|A|I[CR][ETB]8F[CR][LF]
[ACK]
[STX]5R|45|^^^.BACc|+|||A||F||12|||UriSed 3 PRO[CR][ETB]96[CR][LF]
[ACK]
[STX]6C|1|I|A|I[CR][ETB]91[CR][LF]
[ACK]
[STX]7R|46|50235-1^^^MUC|1.5|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]77[CR][LF]
[ACK]
[STX]0R|47|51478-6^^^MUC|6.6|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]89[CR][LF]
[ACK]
[STX]1R|48|53321-6^^^MUC|-|||N||F||12|||UriSed 3 PRO[CR][ETB]93[CR][LF]
[ACK]
[STX]2R|49|^^^AMO|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]0A[CR][LF]
[ACK]
[STX]3R|50|^^^AMO|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]06[CR][LF]
[ACK]
[STX]4R|51|^^^AMO|-|||N||F||12|||UriSed 3 PRO[CR][ETB]27[CR][LF]
[ACK]
[STX]5L|1|N[CR][ETX]08[CR][LF]
[ACK]
[EOT]
```

9.2.1.2 Example 2: AutionMAX (AX-4280) Chemical Sample Measurement Message transfer

```
[ENQ]
[ACK]
[STX]1H|\^&|||AX-4280^AX-4280^2.0.0.0^1^||||||||P|LIS2-
A2|20210706085343[CR][ETB]B9[CR][LF]
[ACK]
[STX]2P|1||||-[CR][ETB]70[CR][LF]
[ACK]
[STX]3O|1|-------------
^G|1^10^12^SAMPLE|C^^^9UB^^|R||||||N|||20210705160500|||||||||||F[CR][ETB]3E[CR][LF]
[ACK]
[STX]4R|1|^^^GLU|50|mg/dl||N||F||-|||AX-4280[CR][ETB]B6[CR][LF]
[ACK]
[STX]5R|2|^^^GLU|+-|||N||F||-|||AX-4280[CR][ETB]D8[CR][LF]
[ACK]
[STX]6R|3|^^^PRO|20|mg/dl||N||F||-|||AX-4280[CR][ETB]C0[CR][LF]
[ACK]
[STX]7R|4|^^^PRO|+-|||N||F||-|||AX-4280[CR][ETB]E5[CR][LF]
[ACK]
[STX]0R|5|^^^BIL||||N||F||-|||AX-4280[CR][ETB]6D[CR][LF]
[ACK]
[STX]1R|6|^^^BIL|-|||N||F||-|||AX-4280[CR][ETB]9C[CR][LF]
[ACK]
[STX]2R|7|^^^URO||||N||F||-|||AX-4280[CR][ETB]90[CR][LF]
[ACK]
[STX]3R|8|^^^URO|NORMAL|||N||F||-|||AX-4280[CR][ETB]5B[CR][LF]
[ACK]
[STX]4R|9|^^^PH|7.5|||N||F||-|||AX-4280[CR][ETB]D0[CR][LF]
[ACK]
[STX]5R|10|^^^PH||||N||F||-|||AX-4280[CR][ETB]5F[CR][LF]
[ACK]
[STX]6R|11|^^^BLD||||N||F||-|||AX-4280[CR][ETB]9B[CR][LF]
[ACK]
[STX]7R|12|^^^BLD|-|||N||F||-|||AX-4280[CR][ETB]CA[CR][LF]
[ACK]
[STX]0R|13|^^^KET|10|mg/dl||A||F||-|||AX-4280[CR][ETB]D0[CR][LF]
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|14|^^^KET|+1|||A||F||-|||AX-4280[CR][ETB]FB[CR][LF]
[ACK]
[STX]3C|1|I|A|I[CR][ETB]8E[CR][LF]
[ACK]
[STX]4R|15|^^^NIT||||A||F||-|||AX-4280[CR][ETB]A9[CR][LF]
[ACK]
[STX]5C|1|I|A|I[CR][ETB]90[CR][LF]
[ACK]
[STX]6R|16|^^^NIT|+2|||A||F||-|||AX-4280[CR][ETB]09[CR][LF]
[ACK]
[STX]7C|1|I|A|I[CR][ETB]92[CR][LF]
[ACK]
[STX]0R|17|^^^LEU|250|Leu/ul||A||F||-|||AX-4280[CR][ETB]6F[CR][LF]
```

```
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|18|^^^LEU||||A||F||-|||AX-4280[CR][ETB]A5[CR][LF]
[ACK]
[STX]3C|1|I|A|I[CR][ETB]8E[CR][LF]
[ACK]
[STX]4R|19|^^^TURB|||N||F||-|||AX-4280[CR][ETB]0C[CR][LF]
[ACK]
[STX]5R|20|^^^TURB|+1|||N||F||-|||AX-4280[CR][ETB]61[CR][LF]
[ACK]
[STX]6R|21|^^^S.G.|1.000|||N||F||-|||AX-4280[CR][ETB]AF[CR][LF]
[ACK]
[STX]7R|22|^^^S.G.||||N||F||-|||AX-4280[CR][ETB]C2[CR][LF]
[ACK]
[STX]0R|23|^^^COLOR|COLORLESS|||N||F||-|||AX-4280[CR][ETB]FB[CR][LF]
[ACK]
[STX]1R|24|^^^COLOR||||N||F||-|||AX-4280[CR][ETB]47[CR][LF]
[ACK]
[STX]2L|1|N[CR][ETX]05[CR][LF]
[ACK]
[EOT]
```

9.2.1.3 Example 3: LabUMat2 Chemical Sample Measurement Message transfer in case of *SI* selected unit

```
[ENQ]
[ACK]
[STX]1H|\^&|||LabUMat2^LabUMat2^3.3.9.3402^1^UPA8101003||||||P|LIS2-
A2|20210715104539[CR][ETB]E9[CR][LF]
[ACK]
[STX]2P|1||||-[CR][ETB]70[CR][LF]
[ACK]
[STX]3O|1|00306_709455045^G|3^6^SuperUser^SAMPLE|C^^^LabStrip U11Plus
GL^0217/7277^20220430235959|R||||||N|||20210707094550|||||||||||F[CR][ETB]70[CR][LF]
[ACK]
[STX]4R|1|68367-2^^^BIL|neg|umol/l||N||F||Operator|||LabUMat2[CR][ETB]AF[CR][LF]
[ACK]
[STX]5R|2|50551-1^^^BIL|neg|||N||F||Operator|||LabUMat2[CR][ETB]4A[CR][LF]
[ACK]
[STX]6R|3|60025-4^^^UBG|35|umol/l||A||F||Operator|||LabUMat2[CR][ETB]CC[CR][LF]
[ACK]
[STX]7C|1|I|A|I[CR][ETB]92[CR][LF]
[ACK]
[STX]0R|4|62487-4^^^UBG|+|||A||F||Operator|||LabUMat2[CR][ETB]40[CR][LF]
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|5|59158-6^^^KET|0.5|mmol/l||A||F||Operator|||LabUMat2[CR][ETB]04[CR][LF]
[ACK]
[STX]3C|1|I|A|I[CR][ETB]8E[CR][LF]
[ACK]
[STX]4R|6|57734-6^^^KET|(+)|||A||F||Operator|||LabUMat2[CR][ETB]9E[CR][LF]
[ACK]
```

```
[STX]5C|1|I|A|I[CR][ETB]90[CR][LF]
[ACK]
[STX]6R|7|5768-7^^^ASC|0.2|g/l||N||F||Operator|||LabUMat2[CR][ETB]88[CR][LF]
[ACK]
[STX]7R|8|1904-2^^^ASC|+|||N||F||Operator|||LabUMat2[CR][ETB]12[CR][LF]
[ACK]
[STX]0R|9|59156-0^^^GLU|norm|mmol/l||N||F||Operator|||LabUMat2[CR][ETB]38[CR][LF]
[ACK]
[STX]1R|10|50555-2^^^GLU|norm|||N||F||Operator|||LabUMat2[CR][ETB]0D[CR][LF]
[ACK]
[STX]2R|11|2888-6^^^PRO|neg|g/l||N||F||Operator|||LabUMat2[CR][ETB]72[CR][LF]
[ACK]
[STX]3R|12|57735-3^^^PRO|neg|||N||F||Operator|||LabUMat2[CR][ETB]A0[CR][LF]
[ACK]
[STX]4R|13|57747-8^^^BLD|neg|Ery/ul||N||F||Operator|||LabUMat2[CR][ETB]CB[CR][LF]
[ACK]
[STX]5R|14|33051-4^^^BLD|neg|||N||F||Operator|||LabUMat2[CR][ETB]77[CR][LF]
[ACK]
[STX]6R|15|50560-2^^^PH|7.5|||N||F||Operator|||LabUMat2[CR][ETB]A1[CR][LF]
[ACK]
[STX]7R|16|50560-2^^^PH|7.5|||N||F||Operator|||LabUMat2[CR][ETB]A3[CR][LF]
[ACK]
[STX]0R|17|20407-3^^^NIT|pos|||A||F||Operator|||LabUMat2[CR][ETB]99[CR][LF]
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|18|50558-6^^^NIT|+|||A||F||Operator|||LabUMat2[CR][ETB]82[CR][LF]
[ACK]
[STX]3C|1|I|A|I[CR][ETB]8E[CR][LF]
[ACK]
[STX]4R|19|58805-3^^^LEU|25|Leu/ul||A||F||Operator|||LabUMat2[CR][ETB]F2[CR][LF]
[ACK]
[STX]5C|1|I|A|I[CR][ETB]90[CR][LF]
[ACK]
[STX]6R|20|60026-2^^^LEU|+|||A||F||Operator|||LabUMat2[CR][ETB]6D[CR][LF]
[ACK]
[STX]7C|1|I|A|I[CR][ETB]92[CR][LF]
[ACK]
[STX]0R|21|53326-5^^^SG|1.001|||N||F||Operator|||LabUMat2[CR][ETB]F6[CR][LF]
[ACK]
[STX]1R|22|^^^SG||||N||F||Operator|||LabUMat2[CR][ETB]A3[CR][LF]
[ACK]
[STX]2R|23|^^^Turbidity||||N||F||Operator|||LabUMat2[CR][ETB]CB[CR][LF]
[ACK]
[STX]3R|24|50552-9^^^Turbidity|Clear|||N||F||Operator|||LabUMat2[CR][ETB]1B[CR][LF]
[ACK]
[STX]4R|25|^^^Color||||N||F||Operator|||LabUMat2[CR][ETB]0E[CR][LF]
[ACK]
[STX]5R|26|50553-7^^^Color|Yellow|||N||F||Operator|||LabUMat2[CR][ETB]F2[CR][LF]
[ACK]
[STX]6L|1|N[CR][ETX]09[CR][LF]
[ACK]
[EOT]
```

9.2.2 QC Measurement Message

QC Message has the same message structure as Sample Measurement Message.

- The differences are in the “sample type” component of the “Instrument Specimen ID” (ISID) field of the test order record: there is “CONTROL” instead of “SAMPLE” in that component. There is ‘Q’ instead of ‘N’ in “Action Code” (AC) field.
- The QC Measurement Message sending is always initiated by the UriSed.
- The sediment QC Measurement Message contains result records only for RBC and WBC parameters.

9.2.2.1 Example 1: Sediment QC Measurement Message transfer

```
[ENQ]
[ACK]
[STX]1H|\^&|||UriSed 3 PRO^UriSed 3 PRO^4.3.45.8513^1^URM07000002||||||Q|LIS2-
A2|20210706122618[CR][ETB]4B[CR][LF]
[ACK]
[STX]2P|1||||QC_LOW[CR][ETB]28[CR][LF]
[ACK]
[STX]3C|1|I|Test passed!|G[CR][ETB]AC[CR][LF]
[ACK]
[STX]4O|1|00106_70612211643^G|1^6^12^CONTROL|S^^^Cuvette^^|R|||||Q|||20210706122116|||||
|||||F[CR][ETB]D6[CR][LF]
[ACK]
[STX]5R|1|46419-8^^^RBC|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]38[CR][LF]
[ACK]
[STX]6R|2|798-9^^^RBC|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]DE[CR][LF]
[ACK]
[STX]7R|3|53292-9^^^RBC|Passed|||N||F||12|||UriSed 3 PRO[CR][ETB]8F[CR][LF]
[ACK]
[STX]0R|4|46702-7^^^WBC|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]35[CR][LF]
[ACK]
[STX]1R|5|51487-7^^^WBC|0.0|p/ul||N||F||12|||UriSed 3 PRO[CR][ETB]40[CR][LF]
[ACK]
[STX]2R|6|53316-6^^^WBC|Passed|||N||F||12|||UriSed 3 PRO[CR][ETB]8C[CR][LF]
[ACK]
[STX]3L|1|N[CR][ETX]06[CR][LF]
[ACK]
[EOT]
```

9.2.2.2 Example 2: Chemical (LabUMat2) QC Measurement Message transfer in case of *Conv.* selected unit

```
[ENQ]
[ACK]
[STX]1H|\^&|||LabUMat2^LabUMat2^3.3.9.3402^1^UPA8101003||||||Q|LIS2-
A2|20210715105026[CR][ETB]E2[CR][LF]
[ACK]
[STX]2P|1||||QC Low[CR][ETB]29[CR][LF]
[ACK]
[STX]3O|1|0121^B|5^5^SuperUser^CONTROL|C^^^LabStrip U11Plus
GL^0217/7277^20220430235959|R|||||Q|||20210707100426||||||||||F[CR][ETB]59[CR][LF]
[ACK]
[STX]4R|1|53327-3^^^BIL|neg|mg/dl||N||F||Service|||LabUMat2[CR][ETB]A6[CR][LF]
```

```
[ACK]
[STX]5R|2|50551-1^^^BIL|neg|||N||F||Service|||LabUMat2[CR][ETB]CF[CR][LF]
[ACK]
[STX]6R|3|50563-6^^^UBG|2|mg/dl||A||F||Service|||LabUMat2[CR][ETB]9E[CR][LF]
[ACK]
[STX]7C|1|I|A|I[CR][ETB]92[CR][LF]
[ACK]
[STX]0R|4|62487-4^^^UBG|+|||A||F||Service|||LabUMat2[CR][ETB]C5[CR][LF]
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|5|50557-8^^^KET|neg|mg/dl||A||F||Service|||LabUMat2[CR][ETB]AF[CR][LF]
[ACK]
[STX]3C|1|I|A|I[CR][ETB]8E[CR][LF]
[ACK]
[STX]4R|6|57734-6^^^KET|neg|||A||F||Service|||LabUMat2[CR][ETB]E1[CR][LF]
[ACK]
[STX]5C|1|I|A|I[CR][ETB]90[CR][LF]
[ACK]
[STX]6R|7|5768-7^^^ASC|20|mg/dl||A||F||Service|||LabUMat2[CR][ETB]A3[CR][LF]
[ACK]
[STX]7C|1|I|A|I[CR][ETB]92[CR][LF]
[ACK]
[STX]0R|8|1904-2^^^ASC|+|||A||F||Service|||LabUMat2[CR][ETB]83[CR][LF]
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|9|53328-1^^^GLU|norm|mg/dl||N||F||Service|||LabUMat2[CR][ETB]3E[CR][LF]
[ACK]
[STX]3R|10|50555-2^^^GLU|norm|||N||F||Service|||LabUMat2[CR][ETB]94[CR][LF]
[ACK]
[STX]4R|11|50561-0^^^PRO|neg|mg/dl||N||F||Service|||LabUMat2[CR][ETB]EB[CR][LF]
[ACK]
[STX]5R|12|57735-3^^^PRO|neg|||N||F||Service|||LabUMat2[CR][ETB]27[CR][LF]
[ACK]
[STX]6R|13|57747-8^^^BLD|neg|Ery/ul||N||F||Service|||LabUMat2[CR][ETB]52[CR][LF]
[ACK]
[STX]7R|14|33051-4^^^BLD|neg|||N||F||Service|||LabUMat2[CR][ETB]FE[CR][LF]
[ACK]
[STX]0R|15|50560-2^^^PH|7.5|||A||F||Service|||LabUMat2[CR][ETB]13[CR][LF]
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|16|50560-2^^^PH|7.5|||A||F||Service|||LabUMat2[CR][ETB]16[CR][LF]
[ACK]
[STX]3C|1|I|A|I[CR][ETB]8E[CR][LF]
[ACK]
[STX]4R|17|20407-3^^^NIT|pos|||A||F||Service|||LabUMat2[CR][ETB]22[CR][LF]
[ACK]
[STX]5C|1|I|A|I[CR][ETB]90[CR][LF]
[ACK]
[STX]6R|18|50558-6^^^NIT|+|||A||F||Service|||LabUMat2[CR][ETB]0B[CR][LF]
[ACK]
[STX]7C|1|I|A|I[CR][ETB]92[CR][LF]
```

```
[ACK]
[STX]0R|19|58805-3^^^LEU|25|Leu/ul||A||F||Service|||LabUMat2[CR][ETB]73[CR][LF]
[ACK]
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
[ACK]
[STX]2R|20|60026-2^^^LEU|+|||A||F||Service|||LabUMat2[CR][ETB]EE[CR][LF]
[ACK]
[STX]3C|1|I|A|I[CR][ETB]8E[CR][LF]
[ACK]
[STX]4R|21|53326-5^^^SG|1.001|||A||F||Service|||LabUMat2[CR][ETB]72[CR][LF]
[ACK]
[STX]5C|1|I|A|I[CR][ETB]90[CR][LF]
[ACK]
[STX]6R|22|^^^SG||||A||F||Service|||LabUMat2[CR][ETB]20[CR][LF]
[ACK]
[STX]7C|1|I|A|I[CR][ETB]92[CR][LF]
[ACK]
[STX]0R|23|^^^Turbidity||||N||F||Service|||LabUMat2[CR][ETB]4E[CR][LF]
[ACK]
[STX]1R|24|50552-9^^^Turbidity|Clear|||N||F||Service|||LabUMat2[CR][ETB]9E[CR][LF]
[ACK]
[STX]2R|25|^^^Color||||N||F||Service|||LabUMat2[CR][ETB]91[CR][LF]
[ACK]
[STX]3R|26|50553-7^^^Color|Yellow|||N||F||Service|||LabUMat2[CR][ETB]75[CR][LF]
[ACK]
[STX]4L|1|N[CR][ETX]07[CR][LF]
[ACK]
[EOT]
```

9.2.3 Result Query Message (load list)

With this message the host is able to query the available results within a date interval. The interval must be defined with the Q record.

9.2.3.1 Example 1: Date Interval Result Query Message

```
[ENQ]
[ACK]
[STX]1H|\^&|||LIS^123^1.0.3|||||||P||20210715132734[CR]Q|1|||||20210705|20210706[CR]L|1|N
[CR][ETX]DB[CR][LF]
[ACK]
[EOT]
```

As a response the analyzer sends the corresponding results to host in the Sample Measurement Message structure.

9.2.4 Worklist Entry Message

This message serves to add worklist entries on the UriSed, received from the host side.

P record contains patient name.

C record is optional. It contains patient comment.

O record contains sample ID, rack and tube positions. 'I' field is fixed in O record. If 'I' contains S or C sample will be measured on sediment or chemical analyser. If 'I' field contains C and S together sample will be measured both analysers. With the I field the Host may give a dilution factor, separated with ^.

9.2.4.1 Example 1: Worklist Entry Messages

```
[ENQ]
[ACK]
[STX]1H|\^&|||LIS^123^1.0.3||||||P||20210721132734[CR]P|1||||Smith[CR]C|1||Blabla|G[CR]O
|1|1234|3^4|CS^3[CR]P|2||||T Pal[CR]O|1|98765|1^1|S[CR]L|1|N[CR][ETX]19[CR][LF]
[ACK]
[EOT]
```

9.2.5 Host Query and Host Query Response Message

The Host Query Message asks the host whether the sample with the given ID should be measured for sediment or not.

According to the host response the UriSed 3 PRO starts to measure or skips the sample. The response can contain patient name, comment and dilution. UriSed 3 PRO will wait up to 40 seconds for the response to a request. If no response is sent or the response is sent by the data manager after 40 seconds, the UriSed 3 PRO will measure the sample.

Note: If for any reason the host does not answer in time with the Host Query Response Message (e.g. network environment is slow) it can result in a decrease of the throughput of the analyzer. In extreme cases it can be advised to disable the host query mode if the speed of the analyzer is important for the customer and the network environment is continuously unstable.

9.2.5.1 Example 1: Host Query and Host Query Response Messages to Measure a Sample

- host query for sample ID 0121 sent from the UriSed 3 PRO
- host query response to sample ID 0121 with patient name, comment and dilution factor are sent to the UriSed 3 PRO to measure this sample
- The bold parts are the arriving data

Query:

```
[ENQ]
[ACK]
[STX]1H|\^&|||UriSed 3 PRO^UriSed 3 PRO^4.3.50.8567^1^URM07000002||||||P|LIS2-
A2|20210721081624[CR]Q|1|^0121^||||||||||O[CR][ETB]7B[CR][LF]
[ACK]
[STX]2L|1|N[CR][ETX]05[CR][LF]
[ACK]
[EOT]
```

Response:

```
[ENQ]
[ACK]
[STX]1H|\^&|||LIS^123^1.0.3||||||P||20210721132734[CR]O|1|0121||^^^1.0|||||||||||||||||
|||Q[CR]L|1|I[CR][ETX]24[CR][LF]
```

```
[ACK]
[EOT]
```

Explanation:

Patient name: **John Smith**

Dilution factor: **3**

Comment: **Comment for John**

The sample with the given ID should be measured for sediment: **Q**

9.2.5.2 Example 2: Host Query Response Messages to Skip a Sample

- host query response to sample ID 0121 is sent to the UriSed 3 PRO to skip this sample
- The bold parts are the arriving data

Query:

```
[ENQ]
[ACK]
[STX]1H|\^&|||UriSed 3 PRO^UriSed 3 PRO^4.3.50.8567^1^URM07000002|||||||P|LIS2-
A2|20210721083039[CR]Q|1|^0121^||||||||||O[CR][ETB]7D[CR][LF]
[ACK]
[STX]2L|1|N[CR][ETX]05[CR][LF]
[ACK]
[EOT]
```

Response:

```
[ENQ]
[ACK]
[STX]1H|\^&|||LIS^123^1.0.3|||||||P||20210721132734[CR]O|1|0121||^^^1.0||||||||||||||||||
|||Y[CR]L|1|I[CR][ETX]2C[CR][LF]
[ACK]
[EOT]
```

Explanation:

The sample with the given ID should be skipped for sediment: **Y**

9.2.6 Sending image

- Image is sent only in case of TCP connection

9.2.6.1 Example 1: Query and response to send an image

Query:

- First the LIS server sends a query to client (UriSed)

```
[ENQ]
[ACK]
```

```
[STX]1H|\^&|||||||||||P|1[CR]M|1|P^3120[CR]Q|1|^004G06_70709505719||||||||||F[CR]
L|1|I[CR][ETX]81[CR][LF]
[ACK]
[EOT]
```

After this through the defined image port the images are sent by the client (UriSed) to LIS server:

Response: Structure of binary data

The transmitted stream looks like this:

- byte 0 to 3: length of fileName in characters (signed integer)
- byte 4 to 7: file length (signed integer)
- byte 8 to n: file name, without ASCII 0 (one character=2 bytes)
- byte n+1 to z: the content of the zip file

Zip file: A zip file is sent to LIS. The name of the zip file is: *"UID_Barcode_DateTime.zip"*. Where:

- **UID** is the unique identifier of the sample in the database
- **Barcode** is the barcode read or generated
- **DateTime** is the date of measurement in yyyymmddhhmmss format.

The contents of the zip file are an images folder and a json file named *"UID_Barcode_DateTime.json"*. Each field in the name has the same meaning as the zip file name.

The "images" folder contains the images that belong to the sample. The images are named "NR_UID_Barcode_DateTime_ImageType.png". Where:

- NR: is the image number in two characters, leading zero.
- ImageType:
  - b: in case of BrightField image
  - p: in case of PhaseContrast image

For example, if the measurement took place at 9:38 on 30/03/2020 and the barcode read is 30011, the database has an ID 247, and there are 15 pairs of images per measurement, then:

Content of 247_30011_202003300938.zip:
  images *(folder)*
    01_247_30011_202003300938_b.png
    01_247_30011_202003300938_p.png
    02_247_30011_202003300938_b.png
    02_247_30011_202003300938_p.png
    …
    15_247_30011_202003300938_b.png
    15_247_30011_202003300938_p.png
  247_30011_202003300938.json *(file)*

The json file has the following structure (field names `field data` *explanation*):

{
„runDateTime":`"yyyy.mm.dd hh:mm:ss"`
„instrumentSerial":`"serialNumber"`
„UID":`unique ID of sample` *this is in filenames*
„imageFormat":`"png"`
„images":

```
[
  {
    „index”:nr image number
    „brFileName”:”brImgName” brightfield name of the image in “images” folder
    „phFileName”:”phcImgName” phcField name of the image in “images” folder (if there is)
    „cells”:
    {
      „RBC”:
      {
        „coordinates”: [[x1,y1], [x2,y2],[x3,y3], … [xn,yn]]
      }
      „WBC”:
      {
        „coordinates”: [[x1,y1], [x2,y2],[x3,y3], … [xn,yn]]
      }
      „CRY”:
      {
        „coordinates”: [[x1,y1], [x2,y2],[x3,y3], … [xn,yn]]
      }
      …
    }
  },
  {
    „index”:nr image number
    „brFileName”:”brImgName” brightField name of the image in “images” folder
    „phFileName”:”phcImgName” phcField name of the image in “images” folder (if there is)
    „cells”:
    {
      „RBC”:
      {
        „coordinates”: [[x1,y1], [x2,y2],[x3,y3], … [xn,yn]]
      }
      „WBC”:
      {
        „coordinates”: [[x1,y1], [x2,y2],[x3,y3], … [xn,yn]]
      }
      „CRY”:
      {
        „coordinates”: [[x1,y1], [x2,y2],[x3,y3], … [xn,yn]]
      }
      …
    }
  },
  …
]
}
```

Important: Since the statistical particles (BAC and its subclasses, MUC, AMO) do not have X, Y coordinates like conventional particles, they do not appear at „cells”.

9.3 Data Record types

9.3.1 Common fields for data records

Abbreviation	RSN
Short explanation	Record Sequence Number
Description	This field is used in record types that may occur multiple times within a single message. The number used defines the ith occurrence of the associated record type at a particular hierarchical level and is reset to one whenever a record of a greater hierarchical significance (lower number) is transmitted or if the same record is used at a different hierarchical level (e.g., comment records)
Example	1

Abbreviation	DT
Short explanation	Date Time codes
Description	Date code should be in YYYYMMDDHHMMSS. Variations for this field: • Date/Time of the Message (DTM) • Date/Time Specimen Received (DTSR) • Date/Time Test Started (DTTS) • Date/Time Test Completed (DTTC) • Expire Date/Time of Operator's information (DTOE)
Example	20190812140130

9.3.2 Message Header Record (H)

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
[STX]	{FN}	H	{DEL}			{SEN}							{MTY}	{PVN}	{DTM}		[CR]	[ETB]	{CHS}	[CR]	[LF]

*From 3…15 all used (including empty) fields start with the field separator (ie.: vertical line character; „|"; ASCII124).

Legend:

(purple)	Fixed fields
(light green)	Common data fields
(yellow)	Unique record fields
(light blue)	Not used fields
(blue)	Frame elements

Example:

```
[STX]1H|\^&|||UriSed 3 PRO^UriSed 3 PRO^4.3.45.8513^1^URM07000002||||||P|LIS2-
A2|20210706085625[CR][ETB]50[CR][LF]
```

Explanation of the fields:

Abbreviation	**DEL**
Short explanation	Delimiters of the protocol
Description	It can be any four ASCII characters except the restricted characters. It is important to use consistently these delimiters in the protocol. The order of the delimiter is: repeat, component, escape. In case of UriSed the content is always the same, static.
Example	\^&

Abbreviation	**SEN**
Short explanation	Sender name or ID
Description	When the UriSed sends this field it contains several components: {Instrument name}^{Software Vendor Organization}^{SW version}^{SW ID}^{Serial Number} • {Instrument name} – UriSed, UriSed 3 PRO • {Software Vendor Organization} – UriSed, UriSed 3 PRO • {SW version} – version of the software • {SW ID} – 1 • {Serial Number} – serial number of the UriSed When the UriSed receives this field as host query response: {SEN} – can be simply empty. The field is not used for special purpose.
Example	UriSed 3 PRO^UriSed 3 PRO^4.3.45.8513^1^URM07000002

Abbreviation	**MTY**
Short explanation	Measurement type
Description	It has got 2 types: • Patient measurement: P • Quality control measurement: Q For host query response: • P: Patient Result
Example	P, Q

Abbreviation	**PVN**
Short explanation	(protocol) Version Number
Description	When the UriSed sends this field it displays the version of the current communication protocol. In this case it should be always „LIS2 A2”. When the UriSed receives this field as host query response: {PVN} – can be simply “1”.
Example	LIS2-A2

Remarks:

- **Frame Number** (FN) in headers is always 1 as this is always the first record in a message.
- **Date/Time of the Message** (DTM) represents the date and time when the message has been sent.

9.3.3 Patient Identifying Record (P)

	0	1	2	3	4	5	6	7				
[STX]	{FN}	P	{RSN}				{PID}	[CR]	[ETB]	{CHS}	[CR]	[LF]

*From 2...6 all used fields start with vertical line character („|”; ASCII124), as field separator.

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields
	Frame elements

#1 Example:

```
[STX]2P|1||||John Smith[CR][ETB]F7[CR][LF]
```

#2 Example (no patient is set):

```
[STX]2P|1||||-[CR][ETB]70[CR][LF]
```

Explanation of the fields:

Abbreviation	**PID**
Short explanation	Patient Name
Description	The Name of the patient will be transmitted. In case of QC measurement QC_HIGH, QC_LOW (sediment) QC High or QC Low (chemical) is sent
Example	Patient Field, QC_LOW

9.3.4 Comment Record (C)

	0	1	2	3	4	5	6				
[STX]	*{FN}*	C	{RSN}	{CTP}	{CTX}	{CTY}	[CR]	[ETB]	{CHS}	[CR]	[LF]

*From 2...5 all used fields start with vertical line character („|”; ASCII124), as field separator.

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields
	Frame elements

Example: result record, followed by a system comment for particle:

```
[STX]1C|1|I|A|I[CR][ETB]8C[CR][LF]
```

Example for user edited comment:

```
[STX]3C|1|I|chemical comment|G[CR][ETB]94[CR][LF]
```

Example for QC system comment:

```
[STX]3C|1|I|Test passed!|G[CR][ETB]AC[CR][LF]
```

Example for query comment:

```
C|1||Comment for John|G[CR]
```

Explanation of the fields:

Abbreviation	**CTP**
Short explanation	Comment Place
Description	System comment, Sediment comment, Particle comment: I Query comment: “”
Example	I

Abbreviation	**CTX**
Short explanation	Comment Text
Description	**System comment:** • “Test passed!” (G) • “Test failed!“(G) • “Not sediment measured” (G) • all the system comments which come from Chemical analyzer **User comment**: Dynamic text, the comment itself given by the user (G) **Sediment status and flags**: • “Empty cuvette” (I) • “Invalid” (I) • “Low level” (I) • “Not measured” (I) • “Dilution factor” (I) • “Review” (I) • “RBC-Aca” (G) • “ghost RBC” (G) System comment for particle (only for Abnormal particles and N/A results): • “A” (I) *Note: abbreviations in () are explained at ‘Comment Type’*
Exampe	This is a comment

Abbreviation	**CTY**
Short explanation	Comment Type
Description	Sediment or chemical user entered comment (and flag names): G System comment (either for sample or particle): I or G according CTX
Example	I

9.3.5 Test Order Record (O)

9.3.5.1 Sending data (sediment result):

	0	1	2	3	4	5	6	7..11	12	13..14	15	16..25	26					
[STX]	*{FN}*	O	{RSN}	{SID}	{ISID}	{UTID}	{TCP}=R		{AC}		{DTTS}		{RT}	[CR]	[ETB]	{CHS}	[CR]	[LF]

*From 2…26 all <u>used</u> fields start with vertical line character („|”; ASCII124), as field separator.

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields
	Frame elements

Example (sample measurement):

```
[STX]3O|1|00210_70515104664^G|2^10^12^SAMPLE|S^^^Cuvette^^|R||||||N|||202
10705151046||||||||||F[CR][ETB]A4[CR][LF]
```

Example (QC measurement):

```
[STX]4O|1|00106_70612211643^G|1^6^12^CONTROL|S^^^Cuvette^^|R||||||Q|||202
10706122116|||||||||||F[CR][ETB]D6[CR][LF]
```

Explanation of the fields:

Abbreviation	**SID**
Short explanation	Specimen ID
Description	Unique ID of the sample. {sampleID}^{idType} • {idType} – type of sample's ID o „B" – code read from sample (barcode) o „G" – generate code Sediment result and host query response too:
#1 Example	00267^B
#2 Example	00106_70612211643^G

Abbreviation	**ISID**
Short explanation	Instrument Specimen ID
Description	{Rack number}^{Tube number}^{User}^{Sample Type} • {Rack number} – the number of the rack maintained by the analyzer • {Tube number} – tube position in the rack • {User} – User name, who is sending the result • {Sample Type} – "SAMPLE" / "CONTROL" ○ "SAMPLE" – marking samples ○ "CONTROL" – for QC measurements
#1 Example	9^5^12^SAMPLE
#2 Example	1^1^12^CONTROL

Abbreviation	**UTID**
Short explanation	Universal Test ID
Description	{Order type}^^^{Name} ^{Lot ID}^{Expiration date} • {Order type} ◦ S – 'S' appears in case of sediment order ◦ C – 'C' appears in case of chemical order • {Name} – Name of consumables • {Lot ID} – Lot ID of consumables • {Expiration date} – Expiration date of consumables
Example	S^^^Cuvette^^

Abbreviation	**TPC**
Short explanation	Test Priority Code
Description	Only R priority code (routine) is allowed.
Example	R

Abbreviation	**AC**
Short explanation	Action Code
Description	{AC} • N – new sample order • Q – quality control order
Example	N

Abbreviation	**RT**
Short explanation	Report type
Description	{RT} • F – Final result (successful measurement) • X – order cannot be done, error occurred
Example	F

Remark: Date/Time Test Started (DTTS) represents the date and time when the test has been started. (see also Common Data Fields 9.3.1)

9.3.5.2 Worklist

1	2	3	4	5	
O	{RSN}	{SID}	{ISID}	{UTID}	[CR]

*From 2…5 all used fields start with vertical line character („|"; ASCII124), as field separator.

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields

Example for sediment without dilution factor:

```
O|1|98765|1^1|S[CR]
```

Example for sediment and chemical with 3 times dilution:

```
O|1|1234|3^4|CS^3[CR]
```

Explanation of the fields:

Abbreviation	**SID**
Short explanation	Specimen ID
Description	Sediment result and host query response too: The unique ID of the sample.
Example	1234

Abbreviation	**ISID**
Short explanation	Instrument Specimen ID
Description	{Rack number}^{Tube number} • {Rack number} – the number of the rack maintained by the analyzer • {Tube number} – tube position in the rack
Example	3^4

Abbreviation	**UTID**
Short explanation	Universal Test ID

Description	{I}^{dilution factor}
	• {I}
	  o S – sediment order
	  o C – chemical order
	• {dilution factor}

Example with dilution factor	CS^3
Example without dilution factor	S

9.3.5.3 Host query response:

1	2	3	4	5	6..25	26	
O	{RSN}	{SID}		{UTID}		{RT}	[CR]

*From 1…26 all used fields start with vertical line character („|”; ASCII124), as field separator.

Legend:

(grey)	Fixed fields
(green)	Common data fields
(yellow)	Unique record fields
(light blue)	Not used fields

Example:

```
O|1|0121||^^^1.0||||||||||||||||||||Q[CR]
```

Explanation of the fields:

Abbreviation	**SID**
Short explanation	Specimen ID
Description	Sediment result and host query response too: The unique ID of the sample.
Example	`0121`

Abbreviation	**UTID**
Short explanation	Universal Test ID
Description	{Not used}^{Not used}^{Not used}^{dilution factor} {dilution factor} – it is the dilution factor for the sample
Example	Not diluted: ^^^1 Dilution factor 3: ^^^3

Abbreviation	**RT**
Short explanation	Report type
Description	{RT} • Q – Response to Query: measure (sent to UriSed as positive response to Host Query) • Y – No order on Record for this test: skip (sent to UriSed as negative response to Host Query)
Example	Q

9.3.6 Result Record (R)

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16			
[STX]	*{FN}*	R	{RSN}	{UTID}	{DMV}	{UNIT}		{RAF}		{RS} = F		{OpID}			{AN}	[CR]	[ETB]	{CHS}	[CR]	[LF]

*From 2…14 all <u>used</u> fields start with vertical line character („|”; ASCII124), as field separator.

Legend:

(lilac)	Fixed fields
(light green)	Common data fields
(yellow)	Unique record fields
(light blue)	Not used fields
(blue)	Frame elements

Example (LabUMat):

```
[STX]6R|3|60025-4^^^UBG|35|umol/l||A||F||Operator|||LabUMat2[CR][ETB]CC[CR][LF]
```

Example (UriSed):

```
[STX]4R|1|46419-8^^^RBC|0.0|p/HPF||N||F||12|||UriSed 3 PRO[CR][ETB]37[CR][LF]
```

Explanation of the fields:

Abbreviation	**UTID**
Short explanation	Universal Test ID
Description	{Test code}^{NU}^{NU}^{Test Name} • {Test code} – Test code (LOINC code) • {NU} – Not used (reserved for internal use) • {Test Name} – name of the tested parameter
Example	46419-8^^^RBC

Abbreviation	**DMV**
Short explanation	Data or Measurement Value
Description	The value (or category) of the measured test
#1 Example	-
#2 Example	+++
#3 Example	0.4

Abbreviation	**UNIT**
Short explanation	Unit
Description	Unit of the result data (in the DMV field). It is always transmitted regardless of the value of the measurement. In case of arbitrary (e.g "++"), this field is empty.
#1 Example	p/ul
#2 Example	p/HPF
#3 Example	mg/dl

Abbreviation	**RAF**
Short explanation	Result Abnormal Flags
Description	If the result was normal: N If the result was abnormal: A
Example	A

Abbreviation	**RS**
Short explanation	Result Status
Description	Only Final results is allowed
Example	F

Abbreviation	**OpID**

Short explanation	Operator Identification / Operator's ID
Description	The ID of the operator which was given when the test was made
Example	Operator

Abbreviation	**AN**
Short explanation	Analyzer Name
Description	Name of the analyzer, static content.
Example	UriSed 3 PRO, LabUMat2

***Sediment* result structure:**

Every particle may get 5 rows

- Exception: if the category result is N/A (then it sends only 1 row/particle)

Sequence of rows	Observation Value	Units
1.	value in p/HPF	p/HPF
2.	value in p/LPF	p/LPF
3.	value in p/ul	p/ul
4.	N°	(empty)
5.	category result (in case of QC: Passed/Failed)	(empty)

***Chemical* result structure in case of *LabUMat2*:**

Every parameter may get 3 rows, except the PMC parameters which have got 1 row

Note: PMC – Physical Measurement Cell- parameters: SG, Turbidity, Color

Sequence of rows	Observation Value	Units
1.	Measured value	Conv.
2.	Measured value	SI
3.	Measured value	(empty)

***Chemical* result structure in case of *AutionMAX*:**

Every parameter has got as many rows as many measure value was in the result.

9.3.7 Message Terminator Record (L)

	0	1	2	3					
[STX]	*{FN}*	L	{RSN}	{TC}	[CR]	[ETB]	{CHS}	[CR]	[LF]

*From 2…15 all <u>used</u> fields start with vertical line character („|"; ASCII124), as field separator.

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields
	Frame elements

Example:

`[STX]2L|1|N[CR][ETX]05[CR][LF]`

Explanation of the fields:

Abbreviation	**TC**
Short explanation	Termination Code
Description	Normal termination: N For host query response: I
Example	N

Remarks:

- **Record sequence number** (RSN) in terminator record is always 1 as this is always the first of this type of record in a message.

9.3.8 Request Information Record (Q)

9.3.8.1 Result query request

1	2	3..6	7	8	
Q	{RSN}		{SD}	{ED}	[CR]

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields

Explanation of the fields:

Abbreviation	**SD**
Short explanation	Start Date
Description	Start date of the requested results
Example	20190203

Explanation of the fields:

Abbreviation	**ED**
Short explanation	End Date
Description	End date of the requested results
Example	20190528

Example:

`Q|1|||||20210705|20210706[CR]`

9.3.8.2 Host query request

1	2	3	4..12	13	
Q	{RSN}	{SID}		{Risc} = O	[CR]

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields

Abbreviation	**SID**
Short explanation	Specimen ID
Description	Unique ID of the sample. ^{sampleID}^{idType}^ • {idType} – type of sample's ID o „G" – code read from sample (barcode)
#1 Example	`^0121^`
#2 Example	`^00405_71512525343^G`

Abbreviation	**RISC**
Short explanation	Request Information Status Code
Description	Only O status code (requesting test orders and demographics only (no result)) is allowed.
Example	O

Example:

```
Q|1|^00406_72013262920^G|||||||||O[CR]
```

9.3.8.3 Picture query request

1	2	3	4..12	13	
Q	{RSN}	{SID}		{RISC} = F	[CR]

Legend:

	Fixed fields
	Common data fields
	Unique record fields
	Not used fields

Example:

```
Q|1| ^004G06_70709505719|||||||||F[CR]
```

Explanation of the fields:

Abbreviation	**SID**
Short explanation	Specimen ID
Description	The unique ID of the sample.
Example	`004G06_70709505719`

Abbreviation	**RISC**
Short explanation	Request Information Status Code
Description	Only Final result is allowed
Example	F

9.3.9 Manufacturer Record (M)

	1	2	3	
	M	{RSN}	P^{IPID}	[CR]

*From 2…3 all used fields start with vertical line character („|”; ASCII124), as field separator.

Legend:

(purple)	Fixed fields
(light green)	Common data fields
(yellow)	Unique record fields
(light blue)	Not used fields

Example:

`M|1|P^3120[CR]`

Explanation of the fields:

Abbreviation	**IPID**
Short explanation	IP Port Number (P in front of the Port Number stands for “port”)
Description	The IP Port Number through the data is be transmitted.
Example	3120

10. HL7 Low level interface

The physical layer of the data exchange is the same as described in TCP/IP protocol standard.

10.1 Data frame elements

This protocol applies the MLLP (Minimum lower layer protocol) over the TCP/IP protocol.

Frame structure:

{SB}	Data content of Message	{EB}	[CR]

Element	Description
{SB}	Start Block character Value: ASCII 0x0B (vertical tab, [VT]) This should not be confused with ASCII characters SOH, STX
{EB}	End Block character Value: 0x1C (file separator, [FS]) This should not be confused with ASCII characters EOT, ETX

11. HL7 High level interface

11.1 Characters and encoding

The UriSed sends and receives strings with ASCII character encoding.

11.1.1 Restricted characters

Restricted characters are the following:

- The start block and end block character. (see above)
- Separators cannot be part of the data of the fields and components. They must be escaped (see below).

The following delimiters are used in the HL7 protocols:

Delimiter type	Name of the character	Sign of the character	ASCII code of the character
Segment terminator	carriage return	[CR]	13
Field separator	vertical bar	|	124
Repeat separator	tilde	~	126
Component separator	caret	^	94
Escape separator	backslash	\	92
Subcomponent separator.	ampersand	&	38

The following escape sequences are defined:

\H\	start highlighting
\N\	normal text (end highlighting)
\F\	field separator
\S\	component separator
\T\	subcomponent separator
\R\	repetition separator
\E\	escape character
\Xdddd...\	hexadecimal data
\Zdddd...\	locally defined escape sequence

11.2 Message types

The following message types exist in HL7 implementation of the UriSed instrument:

- Measurement messages (sent from UriSed to host):
  - Sample Measurement Message
  - QC Measurement Message
  - Host Query Message

- Request messages (sent from host to UriSed):
  - Result Query Message (load list)
  - Worklist Entry Message
  - Host Query Response Message

The examples show the sequence of the communications. The **BLACK** color is the Host and the **GREEN** color is the system (UriSed 3 PRO).

11.2.1 Sample Measurement Message

The measurement message contains the data for a single measurement by the UriSed instrument.

The measurement message sending is always initiated by the UriSed

In case of a connected result the sediment and the chemical results are sent one after another, but the result structures are the same as in case of a standalone result. The result sending order can be set in the user software.

In case of chemical result UriSed sends the result in these units: on chemical analyzer selected unit + the arbitrary value. (It depends on the chemical analyzer type.)

The UriSed uses the following type of message segments header:

MSH-9: "OUL^R22^OUL_R22"

Segments	Description
MSH	Message Header
SFT	Software Segment
PID	Patient Identification
{	--- SPECIMEN begin
SPM	Specimen information
SAC	Container information
{	--- ORDER begin
INV	Inventory Detail Segment
OBR	Observation request
ORC	Common Order
NTE	Notes and Comments
[OBX]	Observation Result(s)
}	--- ORDER end
}	--- SPECIMEN end

11.2.1.1 Sediment Sample Measurement Message transfer

```
[VT]
MSH|^~\&|UriSed 3
PRO^1||||20210719102724||OUL^R22^OUL_R22|20210719102724196|P|2.7|||NE|AL||ASCII[CR]
SFT|UriSed 3 PRO|4.3.50.8567|UriSed 3 PRO|4.3.50.8567[CR]
PID|1||1||-[CR]
```

```
SPM|1|0121&B||UR^Urine^HL70487||||||P^Patient^HL70369[CR]
SAC|||||||||1|9[CR]
INV|Cuvette|OK|||||||||||||[CR]
OBR|1|||UrineSedimentResult[CR]
ORC|RE||0121||CM|||||12|Automatic validated[CR]
OBX|1|ST|46419-8^RBC^LN|1|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|2|ST|798-9^RBC^LN|1|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|3|ST|53292-9^RBC^LN|1|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|4|ST|46702-7^WBC^LN|2|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|5|ST|51487-7^WBC^LN|2|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|6|ST|53316-6^WBC^LN|2|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|7|ST|53307-5^CRY^LN|3|0.2|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|8|ST|53297-8^CRY^LN|3|0.9|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|9|ST|53334-9^CRY^LN|3|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|10|ST|53307-5^.CRY^LN|4|0.2|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|11|ST|53297-8^.CRY^LN|4|0.9|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|12|ST|53334-9^.CRY^LN|4|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|13|ST|^.CaOxm^LN|5|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|14|ST|^.CaOxm^LN|5|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|15|ST|^.CaOxm^LN|5|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|16|ST|^.CaOxd^LN|6|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|17|ST|^.CaOxd^LN|6|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|18|ST|38993-2^.CaOxd^LN|6|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|19|ST|33223-9^HYA^LN|7|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|20|ST|51484-4^HYA^LN|7|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|21|ST|50231-0^HYA^LN|7|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|22|ST|^PAT^LN|8|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|23|ST|^PAT^LN|8|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|24|ST|72224-9^PAT^LN|8|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|25|ST|53294-5^NEC^LN|9|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|26|ST|51485-1^NEC^LN|9|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|27|ST|50225-2^NEC^LN|9|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|28|ST|33219-7^EPI^LN|10|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|29|ST|51486-9^EPI^LN|10|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|30|ST|53318-2^EPI^LN|10|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|31|ST|^YEA^LN|11|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|32|ST|51481-0^YEA^LN|11|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|33|ST|72223-1^YEA^LN|11|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|34|ST|33218-9^BAC^LN|12|10.7|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|35|ST|51480-2^BAC^LN|12|47.1|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|36|ST|50221-1^BAC^LN|12|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|37|ST|33218-9^.BAC^LN|13|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|38|ST|51480-2^.BAC^LN|13|0.0|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|39|ST|50221-1^.BAC^LN|13|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|40|ST|^.BACr^LN|14|1.1|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|41|ST|^.BACr^LN|14|4.8|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|42|ST|^.BACr^LN|14|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|43|ST|^.BACc^LN|15|9.6|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|44|ST|^.BACc^LN|15|42.2|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|45|ST|^.BACc^LN|15|-|||N|||F||||||URM07000002|20210705122222[CR]
OBX|46|ST|50235-1^MUC^LN|16|0.4|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
OBX|47|ST|51478-6^MUC^LN|16|1.8|p/ul||N|||F||||||URM07000002|20210705122222[CR]
OBX|48|ST|53321-6^MUC^LN|16|-|||N|||F||||||URM07000002|20210705122222[CR]
[FS][CR]
```

Host:

```
[VT]
MSH|^~\&|host||cobas 8000||20210719112626||ACK|10001||2.5||||NE||UNICODE UTF-8|[CR]
MSA|AA|20210719102724196||[CR]
[FS][CR]
```

11.2.1.2 AutionMAX (AX-4280) Chemical Sample Measurement Message transfer

```
[VT]
MSH|^~\&|AX-
4280^1||||20210719103535||OUL^R22^OUL_R22|20210719103535157|P|2.7|||NE|AL||ASCII[CR]
SFT|AX-4280|2.0.0.0|AX-4280|0.0.0.0[CR]
```

```
PID|1||1||-[CR]
SPM|1|0121&B||UR^Urine^HL70487||||||P^Patient^HL70369[CR]
SAC|||||||||1|9[CR]
INV|9UB|OK|||||||||||||[CR]
OBR|1|||UrineStripResult[CR]
ORC|RE||0121||CM|||||-|Automatic validated[CR]
OBX|1|ST|^GLU^LN|1|50|mg/dl||N|||F||||||||20210705160500[CR]
OBX|2|ST|^GLU^LN|1|+-|||N|||F||||||||20210705160500[CR]
OBX|3|ST|^PRO^LN|2||||N|||F||||||||20210705160500[CR]
OBX|4|ST|^PRO^LN|2|-|||N|||F||||||||20210705160500[CR]
OBX|5|ST|^BIL^LN|3||||N|||F||||||||20210705160500[CR]
OBX|6|ST|^BIL^LN|3|-|||N|||F||||||||20210705160500[CR]
OBX|7|ST|^URO^LN|4||||N|||F||||||||20210705160500[CR]
OBX|8|ST|^URO^LN|4|NORMAL|||N|||F||||||||20210705160500[CR]
OBX|9|ST|^PH^LN|5|6.5|||N|||F||||||||20210705160500[CR]
OBX|10|ST|^PH^LN|5||||N|||F||||||||20210705160500[CR]
OBX|11|ST|^BLD^LN|6||||N|||F||||||||20210705160500[CR]
OBX|12|ST|^BLD^LN|6|-|||N|||F||||||||20210705160500[CR]
OBX|13|ST|^KET^LN|7|10|mg/dl||A|||F||||||||20210705160500[CR]
OBX|14|ST|^KET^LN|7|+1|||A|||F||||||||20210705160500[CR]
OBX|15|ST|^NIT^LN|8||||A|||F||||||||20210705160500[CR]
OBX|16|ST|^NIT^LN|8|+2|||A|||F||||||||20210705160500[CR]
OBX|17|ST|^LEU^LN|9|75|Leu/ul||A|||F||||||||20210705160500[CR]
OBX|18|ST|^LEU^LN|9||||A|||F||||||||20210705160500[CR]
OBX|19|ST|^TURB^LN|10||||N|||F||||||||20210705160500[CR]
OBX|20|ST|^TURB^LN|10|-|||N|||F||||||||20210705160500[CR]
OBX|21|ST|^S.G.^LN|11|1.000|||N|||F||||||||20210705160500[CR]
OBX|22|ST|^S.G.^LN|11||||N|||F||||||||20210705160500[CR]
OBX|23|ST|^COLOR^LN|12|COLORLESS|||N|||F||||||||20210705160500[CR]
OBX|24|ST|^COLOR^LN|12||||N|||F||||||||20210705160500[CR]
[FS][CR]
```

Host:

```
[VT]
MSH|^~\&|host||cobas 8000||20210719113437||ACK|10002||2.5||||NE||UNICODE UTF-8|[CR]
MSA|AA|20210719103535157||[CR]
[FS][CR]
```

11.2.1.3 LabUMat2 Chemical Sample Measurement Message transfer – in case of selected *SI* unit

```
[VT]
MSH|^~\&|LabUMat2^1||||20210719105723||OUL^R22^OUL_R22|20210719105723834|P|2.7|||NE|AL||A
SCII[CR]
SFT|LabUMat2|3.3.9.3402|LabUMat2|3.3.9.3402[CR]
PID|1||1||-[CR]
SPM|1|00306_709455045&G||UR^Urine^HL70487||||||P^Patient^HL70369[CR]
SAC|||||||||3|6[CR]
INV|LabStrip U11Plus GL|OK||||||||||20220430235959||||0217/7277[CR]
OBR|1|||UrineStripResult[CR]
ORC|RE||00306_709455045||CM|||||Operator|Automatic validated[CR]
OBX|1|ST|68367-2^BIL^LN|1|neg|umol/l||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|2|ST|50551-1^BIL^LN|1|neg|||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|3|ST|60025-4^UBG^LN|2|35|umol/l||A|||F|||||||UPA8101003|20210707094550[CR]
OBX|4|ST|62487-4^UBG^LN|2|+|||A|||F|||||||UPA8101003|20210707094550[CR]
OBX|5|ST|59158-6^KET^LN|3|0.5|mmol/l||A|||F|||||||UPA8101003|20210707094550[CR]
OBX|6|ST|57734-6^KET^LN|3|(+)|||A|||F|||||||UPA8101003|20210707094550[CR]
OBX|7|ST|5768-7^ASC^LN|4|0.2|g/l||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|8|ST|1904-2^ASC^LN|4|+|||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|9|ST|59156-0^GLU^LN|5|norm|mmol/l||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|10|ST|50555-2^GLU^LN|5|norm|||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|11|ST|2888-6^PRO^LN|6|neg|g/l||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|12|ST|57735-3^PRO^LN|6|neg|||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|13|ST|57747-8^BLD^LN|7|neg|Ery/ul||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|14|ST|33051-4^BLD^LN|7|neg|||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|15|ST|50560-2^PH^LN|8|7.5|||N|||F|||||||UPA8101003|20210707094550[CR]
OBX|16|ST|50560-2^PH^LN|8|7.5|||N|||F|||||||UPA8101003|20210707094550[CR]
```

```
OBX|17|ST|20407-3^NIT^LN|9|pos|||A|||F||||||UPA8101003|20210707094550[CR]
OBX|18|ST|50558-6^NIT^LN|9|+|||A|||F||||||UPA8101003|20210707094550[CR]
OBX|19|ST|58805-3^LEU^LN|10|25|Leu/ul||A|||F||||||UPA8101003|20210707094550[CR]
OBX|20|ST|60026-2^LEU^LN|10|+|||A|||F||||||UPA8101003|20210707094550[CR]
OBX|21|ST|53326-5^SG^LN|11|1.001|||N|||F||||||UPA8101003|20210707094550[CR]
OBX|22|ST|^SG^LN|11||||N|||F||||||UPA8101003|20210707094550[CR]
OBX|23|ST|^Turbidity^LN|12||||N|||F||||||UPA8101003|20210707094550[CR]
OBX|24|ST|50552-9^Turbidity^LN|12|Clear|||N|||F||||||UPA8101003|20210707094550[CR]
OBX|25|ST|^Color^LN|13||||N|||F||||||UPA8101003|20210707094550[CR]
OBX|26|ST|50553-7^Color^LN|13|Yellow|||N|||F||||||UPA8101003|20210707094550[CR]
[FS][CR]
```

Host:

```
[VT]
MSH|^~\&|host||cobas 8000||20210719115626||ACK|10004||2.5|||NE||UNICODE UTF-8|[CR]
MSA|AA|20210719105723834||[CR]
[FS][CR]
```

11.2.2 QC Measurement Message

The QC measurement message contains the data for a single QC measurement by the UriSed instrument. The QC measurement message sending is always initiated by the UriSed.

The UriSed uses the following type of message segments header:

MSH-9: “OUL^R22^OUL_R22”

Segments	Description
MSH	Message Header
SFT	Software Segment
PID	
{	--- SPECIMEN begin
SPM	Specimen information
SAC	Container information
{	--- ORDER begin
INV	Inventory Detail Segment
OBR	Observation Order
ORC	Common Order
NTE	Notes and Comments
[OBX]	Observation Result(s)
}	--- ORDER end
}	--- SPECIMEN end

11.2.2.1 Sediment QC Measurement Message transfer

```
[VT]
MSH|^~\&|UriSed 2
(RUS)^1||||20210719120652||OUL^R22^OUL_R22|20210719120652975|P|2.7|||NE|AL||ASCII[CR]
SFT|UriSed 2 (RUS)|4.3.50.8567|UriSed 2 (RUS)|4.3.50.8567[CR]
PID|1||1||QC_LOW[CR]
SPM|1|0121&B||UR^Urine^HL70487|||||||Q^Control specimen^HL70369[CR]
```

```
SAC||||||||||6|5[CR]
INV|Cuvette|OK|||||||||||||[CR]
OBR|1|||UrineSedimentResult[CR]
ORC|RE||0121||CM||||Service|Not yet validated[CR]
NTE|1||Test passed![CR]
OBX|1|ST|46419-8^RBC^LN|1|0.0|p/HPF||N|||F||||||URS04100191|20210707101353[CR]
OBX|2|ST|798-9^RBC^LN|1|0.0|p/ul||N|||F||||||URS04100191|20210707101353[CR]
OBX|3|ST|53292-9^RBC^LN|1|Passed|||N|||F||||||URS04100191|20210707101353[CR]
OBX|4|ST|46702-7^WBC^LN|2|0.0|p/HPF||N|||F||||||URS04100191|20210707101353[CR]
OBX|5|ST|51487-7^WBC^LN|2|0.0|p/ul||N|||F||||||URS04100191|20210707101353[CR]
OBX|6|ST|53316-6^WBC^LN|2|Passed|||N|||F||||||URS04100191|20210707101353[CR]
[FS][CR]
```

Host:

```
[VT]
MSH|^~\&|host||cobas 8000||20210719130554||ACK|10001||2.5||||NE||UNICODE UTF-8|[CR]
MSA|AA|20210719120652975||[CR]
[FS][CR]
```

11.2.2.2 Chemical QC Measurement Message transfer – in case of selected *Conv.* unit

```
[VT]
MSH|^~\&|LabUMat2^1||||20210719121010||OUL^R22^OUL_R22|20210719121010251|P|2.7|||NE|AL||A
SCII[CR]
SFT|LabUMat2|3.3.9.3402|LabUMat2|3.3.9.3402[CR]
PID|1||1||QC High[CR]
SPM|1|00506_710043684&G||UR^Urine^HL70487|||||||Q^Control specimen^HL70369[CR]
SAC||||||||||5|6[CR]
INV|LabStrip U11Plus GL|OK|||||||||20220430235959||||0217/7277[CR]
OBR|1|||UrineStripResult[CR]
ORC|RE||00506_710043684||CM||||Service|Automatic validated[CR]
OBX|1|ST|53327-3^BIL^LN|1|neg|mg/dl||N|||F||||||UPA8101003|20210707100437[CR]
OBX|2|ST|50551-1^BIL^LN|1|neg|||N|||F||||||UPA8101003|20210707100437[CR]
OBX|3|ST|50563-6^UBG^LN|2|2|mg/dl||A|||F||||||UPA8101003|20210707100437[CR]
OBX|4|ST|62487-4^UBG^LN|2|+|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|5|ST|50557-8^KET^LN|3|neg|mg/dl||A|||F||||||UPA8101003|20210707100437[CR]
OBX|6|ST|57734-6^KET^LN|3|neg|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|7|ST|5768-7^ASC^LN|4|20|mg/dl||A|||F||||||UPA8101003|20210707100437[CR]
OBX|8|ST|1904-2^ASC^LN|4|+|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|9|ST|53328-1^GLU^LN|5|norm|mg/dl||A|||F||||||UPA8101003|20210707100437[CR]
OBX|10|ST|50555-2^GLU^LN|5|norm|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|11|ST|50561-0^PRO^LN|6|neg|mg/dl||N|||F||||||UPA8101003|20210707100437[CR]
OBX|12|ST|57735-3^PRO^LN|6|neg|||N|||F||||||UPA8101003|20210707100437[CR]
OBX|13|ST|57747-8^BLD^LN|7|neg|Ery/ul||N|||F||||||UPA8101003|20210707100437[CR]
OBX|14|ST|33051-4^BLD^LN|7|neg|||N|||F||||||UPA8101003|20210707100437[CR]
OBX|15|ST|50560-2^PH^LN|8|7.5|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|16|ST|50560-2^PH^LN|8|7.5|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|17|ST|20407-3^NIT^LN|9|pos|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|18|ST|50558-6^NIT^LN|9|+|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|19|ST|58805-3^LEU^LN|10|25|Leu/ul||A|||F||||||UPA8101003|20210707100437[CR]
OBX|20|ST|60026-2^LEU^LN|10|+|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|21|ST|53326-5^SG^LN|11|1.001|||A|||F||||||UPA8101003|20210707100437[CR]
OBX|22|ST|^SG^LN|11||||A|||F||||||UPA8101003|20210707100437[CR]
OBX|23|ST|^Turbidity^LN|12||||N|||F||||||UPA8101003|20210707100437[CR]
OBX|24|ST|50552-9^Turbidity^LN|12|Clear|||N|||F||||||UPA8101003|20210707100437[CR]
OBX|25|ST|^Color^LN|13||||N|||F||||||UPA8101003|20210707100437[CR]
OBX|26|ST|50553-7^Color^LN|13|Pale yellow|||N|||F||||||UPA8101003|20210707100437[CR]
[FS][CR]
```

Host:

```
[VT]
```

```
MSH|^~\&|host||cobas 8000||20210719130912||ACK|10002||2.5|||NE||UNICODE UTF-8|[CR]
MSA|AA|20210719121010251||[CR]
[FS][CR]
```

11.2.3 Result Query Message (load list)

This message serves to query the available results within a date interval from host side. The interval must be defined with the OBR segment.

The UriSed uses the following type of message segments header:

MSH-9: “OML^O33^OML_O33”

11.2.3.1 Date Interval Result Query Message

```
[VT]
MSH|^~\&|||||20210719132628||OML^O33^OML_O33|11223354|P|2.7[CR]
ORC|NW[CR]
OBR|1||||||20210705|20210707[CR]
[FS][CR]
```

```
[VT]
MSH|^~\&|^UriSed                                                                 3
PRO.1^L||||20210719122727||ACK^O33^ACK_O33|20210719122727154|P|2.7|||NE|NE||ASCII[CR]
MSA|AA|11223354[CR]
[FS][CR]
```

As a response the analyzer sends the corresponding results to host in the Sample Measurement Message structure.

11.2.4 Worklist Entry Message

This message serves to add a worklist entries on the UriSed, received from the host side.

PID contains patient name.

NTE contains general comment for patient after PID.

SPM contains sample ID.

NTE contains chemical(C) and/or sediment(S) order after OBR. It has an optional field with dilution factor. Default is 1 if factor is not used.

The UriSed uses the following type of message segments header:

MSH-9: “OML^O33^OML_O33”

11.2.4.1 Worklist Entry Messages with dilution factor

- with chemical and sediment order and with dilution factor 1.5:

```
[VT]
MSH|^~\&|||||20210719134022||OML^O33^OML_O33|123456|P|2.7[CR]
PID|1||||John Smith[CR]
NTE|1|L|Comment for John|G[CR]
SPM||BAR1122||UR|||||||P[CR]
ORC|NW[CR]
OBR|1[CR]
NTE|1|L|CS^1.5|G[CR]
[FS][CR]
```

```
[VT]
MSH|^~\&|^UriSed 3
PRO.1^L||||20210719124121||ACK^O33^ACK_O33|20210719124121201|P|2.7|||NE|NE||ASCII[CR]
MSA|AA|123456[CR]
[FS][CR]
```

11.2.4.2 Worklist Entry Messages with default dilution factor

- with sediment order only without dilution factor, dilution factor is the default 1:

```
[VT]
MSH|^~\&|||||20210719141913||OML^O33^OML_O33|98765|P|2.7[CR]
PID|1||||Susan Miller[CR]
NTE|1|L|Comment for Susan|G[CR]
SPM||B3344||UR|||||||P[CR]
SAC||||||||||567|3[CR]
ORC|NW[CR]
OBR|1[CR]
NTE|1|L|S|G[CR]
[FS][CR]
```

```
[VT]
MSH|^~\&|^UriSed 3
PRO.1^L||||20210719132012||ACK^O33^ACK_O33|20210719132012176|P|2.7|||NE|NE||ASCII[CR]
MSA|AA|98765[CR]
[FS][CR]
```

11.2.5 Host Query and Host Query Response Message

This message asks the host whether the sample with the given ID should be measured for sediment or not.

According to the host response the UriSed 3 PRO starts to measure or skips the sample. UriSed 3 PRO will wait up to 40 seconds for the response to a request. If no response is sent or the response is sent by the data manager after 40 seconds, the UriSed 3 PRO will measure the sample.

Note: If for any reason the host does not answer in time with the Host Query Response Message (e.g. network environment is slow) it can result in a decrease of the throughput of the analyzer. In extreme cases it can be advised to disable the host query mode if the speed of the analyzer is important for the customer and the network environment is continuously unstable.

11.2.5.1 Example 1: Host Query and Host Query Response Messages to Measure a Sample

- host query for sample ID 0121 sent from the UriSed 3 PRO
- host query response to sample ID 0121 is sent to UriSed 3 PRO to measure this sample
- The bold parts are the arriving data

Query:

```
[VT]
MSH|^~\&|UriSed 3 PRO^1||||20210719144943||QBP^Q11^QBP_Q11|20210719144943282|P|2.7|||NE|NE||ASCII[CR]
QPD|WOS^Work Order Step|IHELAW||0121^[CR]
RCP|I|RD[CR]
[FS][CR]
```

Host Response:

```
[VT]
MSH|^~\&|||||20210719154852||RSP^K11^RSP_K11|11223344|P|2.7|||NE|AL||ASCII[CR]
MSA|AA[CR]
QAK|LaboratoryOrderQuery|OK[CR]
QPD|WOS^Work Order Step^IHELAW||0121[CR]
[FS][CR]
```

```
[VT]
MSH|^~\&|^UriSed 3 PRO.1^L||||20210719144950||ACK^O33^ACK_O33|20210719144950624|P|2.7|||NE|NE||ASCII[CR]
MSA|AA|11223344[CR]
[FS][CR]
```

Explanation:

The sample with the given ID should be *measured* for sediment: **OK**

11.2.5.2 Example 2: Host Query Response Messages to Skip a Sample

- host query response to sample ID 0121 is sent to UriSed 3 PRO to skip this sample
- The bold parts are the arriving data

Response:

```
[VT]
MSH|^~\&|UriSed 3 PRO^1||||20210719145849||QBP^Q11^QBP_Q11|20210719145849975|P|2.7|||NE|NE||ASCII[CR]
QPD|WOS^Work Order Step|IHELAW||0121^[CR]
RCP|I|RD[CR]
[FS][CR]
```

Host Response:

```
[VT]
MSH|^~\&|||||20210719155754||RSP^K11^RSP_K11|11223344|P|2.7|||NE|AL||ASCII[CR]
MSA|AA[CR]
QAK|LaboratoryOrderQuery|AR[CR]
QPD|WOS^Work Order Step^IHELAW||0121[CR]
[FS][CR]
```

```
[VT]
```

```
MSH|^~\&|^UriSed 3
PRO.1^L||||20210719145852||ACK^O33^ACK_O33|20210719145852950|P|2.7|||NE|NE||ASCII[CR]
MSA|AA|11223344[CR]
[FS][CR]
```

The sample with the given ID should be *skipped* for sediment: **AR**

11.3 Control segments

11.3.1 MSH - Message Segment Header

The UriSed uses only the following fields:

MSH-1: |
MSH-2: ^~\&
MSH-3: ^UriSed 3 PRO.1^L
MSH-7: YYYYMMDDHHMMSS
MSH-9: OUL^R22^OUL_R22
MSH-10: YYYYMMDDHHMMSSmmm
MSH-11: P
MSH-12: 2.7
MSH-15: NE
MSH-16: AL
MSH-18: ASCII

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	MSH
1	Field Separator	|
2	Encoding Characters	^~\&
3	Sending Application	**For Host Query:** “” **For Host ACK:** host **For instrument message:** SW short name^HL7 sw modu **For instrument ACK:** ^{SW short name} {HL7 SW modul version}^L (e.g. ^UriSed 3 PRO.1^L)
7	Date/Time of Message	YYYYMMDDHHMMSS (e.g:20210719102724) YYYY: Year MM: Month DD: Days HH: Hours MM: Minutes SS: Seconds
9	Message Type	**For Measured result:** OUL^R22^OUL_R22 **For Host query message:** OML^O33^OML_O33 **For System query message:**

		QBP^Q11^QBP_Q11 **For Host query response message:** RSP^K11^RSP_K11
10	Message Control ID	{uniqueID} (uniquely identifies the message)
11	Processing ID	"P"
12	HL7 version	"2.7"
15	Accept Acknowledgment Type	"NE"
16	Application Acknowledgment Type	The sender waits for ACK response: "AL" The sender don't wait for ACK response: "NE"
18	Character Set	"ASCII"

11.3.1.1 Example:

```
MSH|^~\&|UriSed 3 PRO^1||||20210719102724||OUL^R22^OUL_R22|20210719102724196|P|2.7|||NE|AL||ASCII[CR]
```

11.3.2 SFT - Software segment

The UriSed uses only the following fields:

SFT-1: UriSed
SFT-2: 4.1.65.7325
SFT-3: UriSed
SFT-4: 4.1.65.7325

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	SFT
1	Software Vendor Organization	UriSed or LabUMat2 or AutionMax (Instrument name)
2	Software Certified Version or Release Number	4.3.50.8567 or chemical analyzer's software version
3	Software Product Name	UriSed or LabUMat2 or AutionMax (Instrument name)
4	Software Binary ID	4.3.50.8567 or chemical analyzer's software version

11.3.2.1 Example:

```
SFT|UriSed 3 PRO|4.3.50.8567|UriSed 3 PRO|4.3.50.8567[CR]
```

11.4 Data segments

11.4.1 PID - Patient Identifying Segment

The UriSed uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	PID
1	Set ID - PID	1
3	Patient Identifier List	1
5	Patient Name	Patient Name or QC lot level

Example for worklist:

`PID|1||||John Smith[CR]`

Example for QC:

`PID|1||1||QC_LOW[CR]`

Example for result without name:

`PID|1||1||-[CR]`

11.4.2 SPM - Specimen Segment

The UriSed uses only the following fields:

11.4.2.1 UriSed sending in a measurement message:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	SPM
1	Set ID – SPM	1
2	Specimen ID	sampleID&idType - {idType} – type of sample's ID ○ „B" – code read from sample (barcode) ○ „G" – generate code
4	Specimen Type	UR^Urine^HL70487
11	Specimen Role	Sample: P^Patient^HL70369 QC: Q^Control·specimen^HL70369

Example:

`SPM|1|0121&B||UR^Urine^HL70487||||||P^Patient^HL70369[CR`

Example QC:

`SPM|1|00506_710043684&G||UR^Urine^HL70487||||||Q^Control specimen^HL70369[CR]`

11.4.2.2 UriSed receiving worklist

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	SPM
2	Specimen ID	Sample ID
4	Specimen Type	UR
11	Specimen Role	P

Example:

```
SPM||BAR1122||UR|||||||P[CR]
```

11.4.3 SAC - Container information Segment

The UriSed uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	SAC
10	Carrier Identifier	Rack number
11	Position in Carrier	Tube position

Example:

```
SAC||||||||||1|9[CR]
```

11.4.4 INV – Inventory Detail Segment

Segment provided when a chemical result is transmit.

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	INV
1	Substance Identifier	**For sediment:** Cuvette **For AutionMAX:** 9UV **For LabUMat2:** LabStrip U11PLUS GL
2	Substance Status	OK
12	Expiration Date/Time	"YYYYMMDDHHMMSS"
16	Manufacturer Lot Number	RG325017

Example for Chemical:

```
INV|LabStrip U11Plus GL|OK|||||||||20220430235959||||0217/7277[CR]
```

Example for Sediment:

```
INV|Cuvette|OK||||||||||||||[CR]
```

```
The 12. and 16. sequent is empty. The AutionMax is same.
```

11.4.5 OBR - Observation request segment

The UriSed uses only the following fields:

11.4.5.1 UriSed sending in a measurement message:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	OBR
1	Set ID – OBR	1
4	Universal Service Identifier	**For Sediment result:** UrineSedimentResult **For Chemical result:** UrineStripResult

Example:

```
OBR|1|||UrineSedimentResult[CR]
```

11.4.5.2 UriSed receiving result query message (load list):

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	OBR
1	Set ID – OBR	1
7	Observation Date/Time	20190601
8	Observation End Date/Time	20190606

Example:

```
OBR|1||||||20210705|20210707[CR]
```

11.4.5.3 UriSed receiving worklist query message:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	OBR
1	Set ID – OBR	1

Example:

```
OBR|1[CR]
```

11.4.6 ORC - Common order

The UriSed uses only the following fields:

11.4.6.1 UriSed sending in a measurement message:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	ORC
1	Order Control	RE
3	Filler Order Number	Sample ID
5	Order Status	CM (measurement successful) ER (measurement error)

10	Entered By	user name (at measure start)
11	Verified By	**by User**: user name (at validation) **by Urised:** Automatic validated **Not verified:** Not yet validated

Example:

`ORC|RE||0121||CM||||Operator|SuperUser[CR]`

11.4.6.2 UriSed receiving a result query (load list) or a worklist entry message:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	ORC
1	Order Control	NW

Example:

`ORC|NW[CR]`

11.4.7 NTE - notes and comments segment

The UriSed uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	NTE
1	Set ID - NTE	Comment sequence
2	Source of Comment	System/User comment: "" empty Worklist query: L
3	Comment	Comment text
4	Comment Type	"RE" (remark) for system comment messages. "G" for General Instruction Not used for user entered comment, for "Not sediment measured" comment, and QC result (Test passed!/Test failed!) system comments.

Example for worklist query with 1.5 dilution factor:

`NTE|1|L|CS^1.5|G[CR]`

Example for QC case:

`NTE|1||Test passed![CR]`

11.4.8 OBX - Observation/Result Segment

The UriSed uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	OBX
1	Set ID – OBX	Sequence number, starting with 1
2	Value Type	“ST”
3	Observation Identifier	LOINC number(value)^ Description^LN
4	Observation Sub-ID	Parameter sequence number
5	Observation Value	Result value
6	Units	Result unit
8	Interpretation Codes	N: Normal A: Abnormal (N/A results are abnormal)
11	Observation Result Status	“F”
18	Equipment Instance Identifier	Serial number
19	Date/Time of the Observation	Measurement DateTime YYYYMMDDHHMMSS

Structure of *sediment* OBX segment:

Every particle may get 5 rows

- Exception: if the category result is N/A (then it sends only 1 row/particle)

Sequence of rows	Observation Value	Units
1.	value in p/HPF	p/HPF
2.	value in p/LPF	p/LPF
3.	value in p/ul	p/ul
4.	N°	(empty)
5.	category result (in case of QC: Passed/Failed)	(empty)

Example for one particle result:

```
OBX|1|ST|46419-8^RBC^LN|1|0.0|p/HPF||N|||F||||||URM07000002|20210705122222[CR]
```

Example QC result - Passed:

```
OBX|6|ST|53316-6^WBC^LN|2|Passed|||N|||F||||||URS04100191|20210707101353[CR]
```

Structure of *chemical* OBX segment in case of *LabUMat2*:

Every parameter may get 3 rows, except the PMC parameters which have got 1 row

Note: PMC – Physical Measurement Cell - parameters: SG, Turbidity, Color

The turbidity values: Clear, Light Turbid, Very Turbid, (in case of ‘No parameter’ this row is not sent)

Sequence of rows	Observation Value	Units
1.	Measured value	Conv.
2.	Measured value	SI
3.	Measured value	(empty)

Structure of *chemical* OBX segment in case of *AutionMAX*:

Every parameter has got as many rows as many measured values were in the result.

Example chemical result in case of AutionMAX:

```
OBX|16|ST|^NIT^LN|8|+2|||A|||F||||||||20210705160500[CR]
```

11.5 Host Communication segments

11.5.1 MSA – Message Acknowledgement Segment

The Host and the UriSed uses the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	MSA
1	Acknowledgment Code	HL7 ack table(AA, AR)
2	Message Control ID	ID of the recived data

Example for ACK codes:

AA: Application Accept

AE: Application Error

AR: Application Reject

Example for system answer:

`MSA|AA|11223354[CR]`

Example for Host answer:

`MSA|AA|20210719121010251||[CR]`

11.5.2 QAK – Query Acknowledgement Segment

The Host uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	QAK
1	Query Tag	LaboratoryOrderQuery
2	Query Response Status	According the HL7 protocol (OK, AR, AE)

Example when data found and no error occur:

`QAK|LaboratoryOrderQuery|OK`

Example with application error:

`QAK|LaboratoryOrderQuery|AR`

11.5.3 QPD – Query Parameter Definition

The Host uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	QPD
1	Message Querry Name	WOS^Work·Order·Step^IHELAW
3	Sample ID	Response with ID

Example:
`QPD|WOS^Work·Order·Step^IHELAW||0121`

The UriSed uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	QPD
1	Message Querry Name	WOS^Work Ord0er Step
2	Query Tag	IHELAW
4	Sample ID	**Normal ID:** Actual sample ID^ **Generated ID:** Actual sample ID^G

Example:
`QPD|WOS^Work Ord0er Step|IHELAW||0121^`

Example with generated Sample ID:
`QPD|WOS^Work Ord0er Step|IHELAW||00306_709455045^G`

11.5.4 RCP - Response Control Parameter

The UriSed uses only the following fields:

SEQ	ELEMENT NAME	Local application/ implementation
0	Segment Identifier	RCP
1	Query Priority	I (Immediate)
2	Quantity Limited Request	RD (Records)

Example:
`RCP|I|RD[CR]`

12. Particle LOINC codes

Class	particle	results	LOINC#
Red blood cells	RBC	category	53292-9
		p/HPF	46419-8
		p/uL	798-9
White blood cells	WBC	category	53316-6
		p/HPF	46702-7
		p/uL	51487-7
	WBCc	category	53317-4
		p/HPF	50233-6
		p/uL	33768-3
Squamous Epithelial Cells	EPI	category	53318-2
		p/HPF	33219-7
		p/uL	51486-9
Non Squamous Epithelial Cells	NEC	category	50225-2
		p/HPF	53294-5
		p/uL	51485-1
	s-TRA	category	
		p/HPF	
		p/uL	
	d-TRA	category	
		p/HPF	
		p/uL	
	REN	category	53271-3
		p/HPF	33221-3
		p/uL	53274-7
Lipids	LIP	category	
		p/HPF	
		p/uL	
	REN-L	category	50228-6
		p/HPF	53354-7
		p/uL	53352-1
	LDR	category	-
		p/HPF	-
		p/uL	-
	CHOL	category	50223-7
		p/HPF	55367-7
		p/uL	-
Hyaline casts	HYA	category	50231-0
		p/HPF	33223-9
		p/uL	51484-4
Pathological casts	PAT	category	72224-9
		p/HPF	-
		p/uL	-
	C-HGR	category	-
		p/HPF	-

		p/uL	-
	C-GRA	category	50230-2
		p/HPF	33341-9
		p/uL	53282-0
	C-NEC	category	50224-5
		p/HPF	53291-1
		p/uL	53287-9
	C-RBC	category	53278-8
		p/HPF	33229-6
		p/uL	53285-3
	C-WBC	category	53279-6
		p/HPF	33228-8
		p/uL	53286-1
	C-CRY	category	-
		p/HPF	-
		p/uL	-
	C-MIC	category	-
		p/HPF	-
		p/uL	-
	C-FAT	category	50229-4
		p/HPF	33231-2
		p/uL	53288-7
	C-WAX	category	41190-0
		p/HPF	33230-4
		p/uL	41868-1
	C-MIX	category	50234-4
		p/HPF	53289-5
		p/uL	53283-8
Crystals	CRY	category	53334-9
		p/HPF	53307-5
		p/uL	53297-8
	CaOx	category	33234-6
		p/HPF	53306-7
		p/uL	53296-0
	CaOxm	category	-
		p/HPF	-
		p/uL	-
	CaOxd	category	38993-2
		p/HPF	-
		p/uL	-
	TRI	category	33238-7
		p/HPF	53308-3
		p/uL	53298-6
	URI	category	33233-8
		p/HPF	53311-7
		p/uL	53301-8
	cApH	category	33235-3
		p/HPF	53309-1
		p/uL	53299-4
	U-AMO	category	50239-3

		p/HPF	32150-5
		p/uL	13657-2
	P-AMO	category	50236-9
		p/HPF	55378-4
		p/uL	13656-4
	CYS	category	33240-3
		p/HPF	53313-3
		p/uL	53303-4
	LEU	category	50232-8
		p/HPF	53310-9
		p/uL	53300-0
	TYR	category	50238-5
		p/HPF	53314-1
		p/uL	53304-2
	ATY	category	53329-9
		p/HPF	53332-3
		p/uL	53331-5
Yeast	YEA	category	72223-1
		p/HPF	-
		p/uL	51481-0
Bacteria	BAC	category	50221-1
		p/HPF	*33218-9*
		p/uL	*51480-2*
	BACr	category	-
		p/HPF	-
		p/uL	-
	BACc	category	-
		p/HPF	-
		p/uL	-
Mucus	MUC	category	53321-6
		p/HPF	*50235-1*
		p/uL	*51478-6*
Spermatozoa	SPRM	category	33232-0
		p/HPF	53324-0
		p/uL	51479-4
Trichomonas	TRV	category	50237-7
		p/HPF	53357-0
		p/uL	53355-4
Parasites	SCH	category	33017-5
		p/HPF	-
		p/uL	-
Artifacts	ART	category	42578-5
		p/HPF	-
		p/uL	-

13. Definitions and acronyms

Below are the definitions of terms used within this document.

Term	Definition
LIS	Laboratory Informational System.
Host	In this document it refers to the laboratory informational system. It communicates with the UriSed and with other network elements.
LAN	Local Area Network.
RS-232	Serial connection standard EIA RS-232.
USB	Universal Serial Bus.
Ethernet	IEEE 802.3 standard connection.
UriSed	Urine analyzer refers to the laboratory instrument. It communicates with the HOST.
Message	A series of records of the same purpose.
Record	One line of transmission closed with [CR][LF].
Field	This is the sub-element of the record separated by field separator character.
Component	This is the sub-element of the field separated with the component separator character.

14. Modification History

Version	Date (dd.mm.yyyy)	Made by	**Modifications** (description of modifications; effected documents, trainings & other activities needed to entry into force of the document)
4	23.07.2021	Csaba Tusa	-All example regenerates. -The example font type is changed to the “Courier New” monospaced serif typeface. - 4.2.1. New lines about the result displays exclusively in SI or Conv unit. The structure of the frame is updated. -5.3.2.1., 6.3.4.1. and 7.3.4.1. The frame gets ^Reserved11^Reserved12/^Reserved1N1^ReservedN2 between the Unit11/Unit1N and Result12/ResultN2. The Unit12/UnitN2 part is rewrite to Reserved13/ ReservedN3 -5.3.2.1. The frame names of result parts rewritten: The result3 was written to result2 and the original result2 become Reserved1. The frame names of unit parts rewritten: The unit2 was written to Reserved2 and the unit3 become Reserved3. -5.3.2.3 and 7.3.4.3. Redundant tables are deleted. -5.3.2.5., 6.3.4.5. and 7.3.4.5. Unused rows rewrite to “Reserved” in the structure of the results table. A new line about the Reserved part is added/updated. -5.3.2.2., 6.3.4.2. and 7.3.4.2. The unnecessary column of ResultN2 is deleted and the ResultN3 is renamed to ResultN2. -9.2.1. The third and the fifth footnote are made complete. And the fourth footnote rewrote to “Test Order Record (O)” from “Manufacturer Record”. -6.3.4. Write a small paragraph about both the chemical and sediment result is sending in the case of chemical result sending. -5.3.1.4 and 7.3.3.4. the QC_LEVEL -1 results are rewrite to “-1 or 255” and get a new “The value depends on the host decoding method.” line. -6.3.3.4 and 7.3.3.4 the table of the “structure of result” replaced with the right table. - 9.2.1.3. An unnecessary table is deleted. -5.3 and 7.3 The RESERVE is rewrite to RESERVED. -5.3.1.4., 5.3.2.5., 7.3.3.4. and 7.3.4.5. The Sample Status is corrected. -9.3.5.2 and 9.3.5.3. The field structure is corrected. -9.3.6. and 11.4.8. The table of sediment and LabuMat2 result structure is improved. The descriptions are rewrite from “has got” to “may get”. -9.3.8 Get a new sub part “Host query”. The sub parts are renamed lost the “In case” part and get a “request” on the end. The Host query get the 9.3.8.2. number and the picture query become the 9.3.8.3. -11.2.1. and 11.2.2. Add a new INV line to the message segment table. - 11.3.1. In the MSH-9 structure table the message types are corrected. -11.4.1., 11.4.4., 11.4.5.1. and 11.4.5.2. The table of measurement message is corrected. -11.4.5.3. a new case of ORB is added. 11.3. and 11.4. all the segment table get a segment identifier row.

			-9.3.4. The query comment examples are added, the field descriptions are corrected. -11. Added a new Host Communication segments part (11.5) -11.4.6.1. and 11.4.7. structure tables of the messages are corrected. - The particle LONIC codes becomes the 12. main chapter. -9.3. Correct the visualization of the frame structure to all LIS A2 data record type.
3	22.01.2021	Brigitta Krisztina Losonczi	- Examples for the LIS and HL7 protocols are updated because of barcode information implementation. - 5.3.2.1, 6.3.4.1, 7.3.4.1, 9.2.1, 11.2.1 (frame in case of normal and QC chemical results) are updated. - Examples updating for the following subchapters: 5.3.2.3, 5.3.2.4; 6.3.4.3, 6.3.4.4; 7.3.4.3, 7.3.4.4; 9.2.1.3, 9.2.2.2; 11.2.1.3, 11.2.2.2 - 9.3.9.1 In case of only result query section is modified with Start Date field. is updated. - 9.2.6 Sending image section, 9.3.4 Manufacturer Record (M) and - 9.3.9.2 In case of picture query sections are added. - 7.3.4.5 Delete duplication: “RESERVE N: This part of the frame is reserved for future usage.” -7.3.4.5 Adding the missing “MEAS_TYPE” -7.3.4.5 Correct the “SG_VALID” flag values. - LOINC codes are updated in the 11th chapther.
2	04.09.2019	Karman Kartosonto	- Deletion “5.3.1.2 Structure of the Results” - Adding columns “ResultN1”, “ResultN2” to chemical result tables where it was necessary, e.g. 5.3.2.2, 7.3.4.2 - Deletion “non-PHC1 instrument” from the subchapters’ titles, e.g. 6.3.3.2, 6.3.3.3, 7.3.3.2, 7.3.3.3 - Adding new chemical results to subchapters 6.3.4.2-6.3.4.4 - Writing “p/HPF” and “p/ul” Result record units in the correct order in subchapter 9.2.1 - Deletion of incorrect notes in subchapter 9.2.1.3 - Improving chemical (LabUMat2) QC examples in subchapter 9.2.2.2 - Separating “Chemical result structures” in case of LabUMat2 and AutionMAX in subchapter 9.3.5 - Improving “LabUMat2 Chemical Sample Measurement Message transfer” example in subchapter 11.2.1.3 - Adding “Structure of chemical OBX segment” in case of LabUMat2 and AutionMAX to subchapter 11.4.7
1	30.08.2019	Karman Kartosonto	First release.